

Annual Review of Virology

CBASS to cGAS-STING: The Origins and Mechanisms of Nucleotide Second Messenger Immune Signaling

Kailey M. Slavik^{1,2} and Philip J. Kranzusch^{1,2,3}

¹Department of Microbiology, Harvard Medical School, Boston, Massachusetts, USA; email: philip_kranzusch@dfci.harvard.edu

²Department of Cancer Immunology and Virology, Dana-Farber Cancer Institute, Boston, Massachusetts, USA

³Parker Institute for Cancer Immunotherapy at Dana-Farber Cancer Institute, Boston, Massachusetts, USA

Annu. Rev. Virol. 2023. 10:423–53

First published as a Review in Advance on June 28, 2023

The *Annual Review of Virology* is online at virology.annualreviews.org

<https://doi.org/10.1146/annurev-virology-111821-115636>

Copyright © 2023 by the author(s). This work is licensed under a Creative Commons Attribution 4.0 International License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See credit lines of images or other third-party material in this article for license information.

ANNUAL REVIEWS CONNECT

www.annualreviews.org

- Download figures
- Navigate cited references
- Keyword search
- Explore related articles
- Share via email or social media

Keywords

cGAS, STING, cGLR, CBASS, nucleotide second messenger, antiviral defense, innate immunity

Abstract

Host defense against viral pathogens is an essential function for all living organisms. In cell-intrinsic innate immunity, dedicated sensor proteins recognize molecular signatures of infection and communicate to downstream adaptor or effector proteins to activate immune defense. Remarkably, recent evidence demonstrates that much of the core machinery of innate immunity is shared across eukaryotic and prokaryotic domains of life. Here, we review a pioneering example of evolutionary conservation in innate immunity: the animal cGAS-STING (cyclic GMP-AMP synthase–stimulator of interferon genes) signaling pathway and its ancestor in bacteria, CBASS (cyclic nucleotide-based antiphage signaling system) antiphage defense. We discuss the unique mechanism by which animal cGLRs (cGAS-like receptors) and bacterial CD-NTases (cGAS/dinucleotide-cyclase in *Vibrio* (DncV)-like nucleotidyltransferases) in these pathways link pathogen detection with immune activation using nucleotide second messenger signals. Comparing the biochemical, structural, and mechanistic details of cGAS-STING, cGLR signaling, and CBASS, we highlight emerging questions in the field and

examine evolutionary pressures that may have shaped the origins of nucleotide second messenger signaling in antiviral defense.

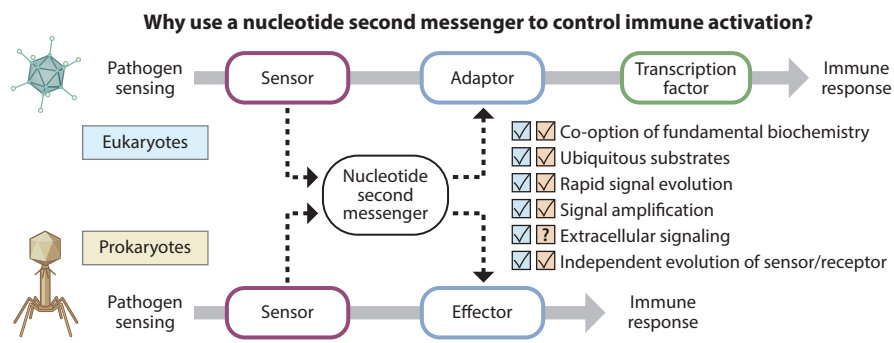
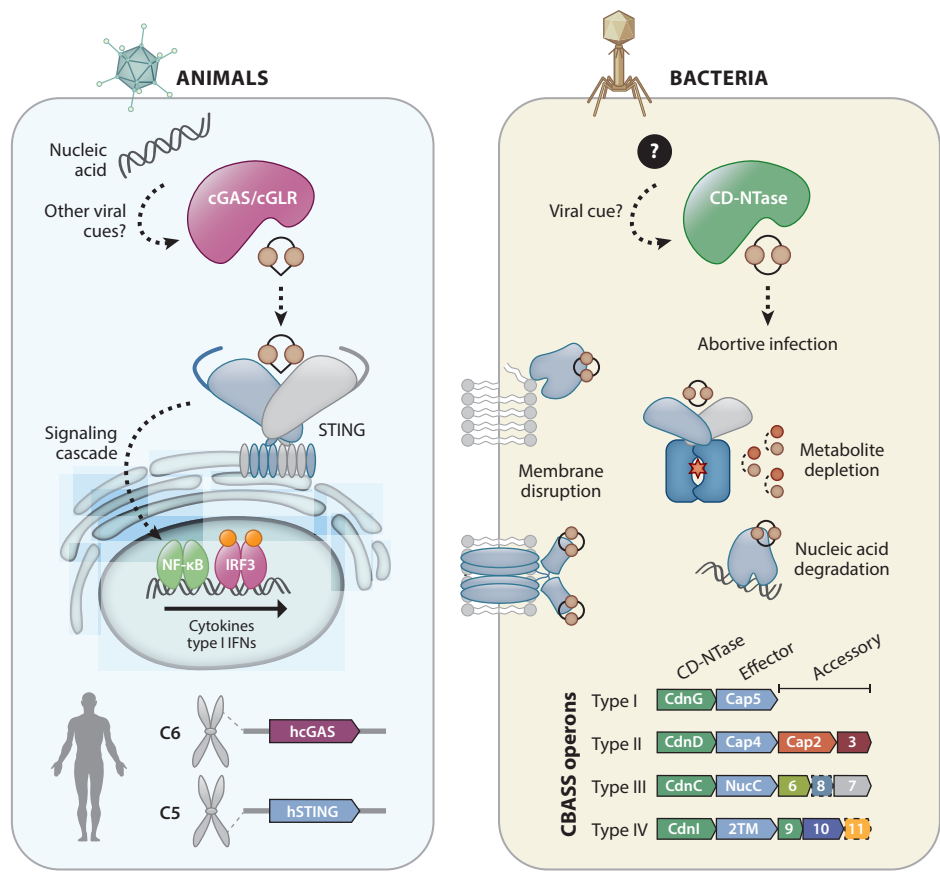
1. INTRODUCTION

cGAS-STING (cyclic GMP-AMP synthase–stimulator of interferon genes) is an immune pathway in animal cells that controls the cellular response to cytosolic double-stranded (ds) DNA (1–3). Upon directly binding to dsDNA, the enzyme cGAS (cyclic GMP-AMP synthase) is activated to synthesize the nucleotide second messenger 2'3'-cGAMP (cG[2'–5']pA[3'–5']p) (4–7). 2'3'-cGAMP is recognized by the adaptor protein STING (stimulator of interferon genes), which then recruits the kinase TBK1 (TANK-binding kinase 1) and the transcription factors IRF3 (interferon regulatory factor 3) and NF-κB (nuclear factor κ-light-chain-enhancer of activated B cells) to activate type I IFN (interferon) and an antiviral signaling program (8). Most animal species encode multiple homologs of cGAS, known as cGLRs (cGAS-like receptors) (9, 10). The best characterized of these, *Drosophila melanogaster* cGLR1, functions as a dsRNA sensor and synthesizes a distinct nucleotide signal, 3'2'-cGAMP, to activate STING and antiviral immunity (9, 10). Divergent cGLRs encoded throughout the animal kingdom recognize dsDNA, dsRNA, or as-yet-unidentified ligands and synthesize a variety of nucleotide second messengers that bind STING proteins encoded within corresponding species (11). Thus, cGLR-STING signaling is a broadly conserved mechanism in metazoan innate immunity that enables animals to detect diverse molecular cues of pathogen infection and use nucleotide signals to control the immune response (**Figure 1**).

Identification of ancient cGAS and STING proteins in bacteria led to the striking discovery that many metazoan innate immune pathways first evolved in prokaryotes as antibacteriophage (antiphage) defense systems (12–14). Bacterial cGAS homologs, known as CD-NTases (cGAS/DncV-like nucleotidyltransferases), control a form of antiphage defense named CBASS (cyclic oligonucleotide-based antiphage signaling system) that is present in an estimated ~14.4% of sequenced bacterial and archaeal species (12, 15, 16). Similar to cGAS recognition of viral infection in animal cells, bacterial CD-NTases sense phage infection and synthesize a diversity of nucleotide signals to activate Cap (CD-NTase-associated protein) effectors (13, 17, 18). CBASS systems encode diverse Cap effectors that upon activation degrade essential cellular components to induce abortive infection, thereby inhibiting the production of progeny phage particles within the infected cell and limiting viral spread (13, 17–19) (**Figure 1**). CBASS systems often include ancillary Cap proteins that regulate CD-NTase activation, and systems can be stratified into four types based on conserved operon structure (for an explanation, see the caption to **Figure 1**) (15).

In many classical forms of innate immunity, protein sensors detect molecules associated with infection and then transmit immune activation through direct protein–protein interactions (**Figure 1**). Hallmark examples include mammalian Toll-like receptors that directly bind adaptor proteins to control downstream signal transduction and the transcription of antiviral genes (20, 21). Similarly, many bacterial defense systems function through direct protein–protein interactions to form complexes that both sense and degrade target phage molecules (19, 22, 23). cGLR-STING and CBASS immunity are two systems that subvert this paradigm: Instead of protein–protein interactions, immune activation is communicated via the production and reception of a diffusible nucleotide second messenger. Additional forms of immunity using nucleotide second messengers, including Type III CRISPR (24, 25), Pycsar (26), and Thoeris (27, 28) antiphage defense systems in bacteria and OAS (oligoadenylate synthase) enzymes (29) in animal cells, highlight the central evolutionary role of second messenger signals in immune defense.

Here we compare the biochemical, structural, and mechanistic details of metazoan cGAS-STING and bacterial CBASS signaling to explore the role of nucleotide signals in antiviral immunity and how the evolution of ancient pathways in bacteria shaped animal antiviral defenses. Major questions include, What makes these forms of immunity unique? What advantage does this type of signaling confer to immune systems in metazoans and bacteria? How have viruses evolved to subvert such pathways? Here we discuss each of these open questions and highlight how the



(Caption appears on following page)

Figure 1 (Figure appears on preceding page)

Nucleotide second messenger signaling in cGLR-STING and CBASS immunity. In animals, cGAS and cGLR pattern recognition receptors detect molecular signatures of infection and synthesize cyclic dinucleotide second messengers. Upon binding nucleotide second messenger signals, the adaptor protein STING activates antiviral transcription programs through NF- κ B and IRF3. cGLR enzymes and STING are encoded at separate genomic loci: In humans cGAS is encoded on C6, and STING is encoded on C5. In bacterial CBASS systems, CD-NTase enzymes are activated by phage infection to synthesize cyclic di- or trinucleotide second messengers that are specifically recognized by adjacently encoded Cap effectors. Cap effectors target essential cellular components to induce abortive infection and protect bacterial populations from phage infection. CBASS operons universally encode a CD-NTase and an effector protein and are classified by regulatory Cap proteins. Type I systems contain no additional proteins. Type II systems encode Cap2, a protein with homology to eukaryotic ubiquitin-activating E1 proteins and ubiquitin-conjugating E2 proteins, and Cap3, a protein with homology to eukaryotic JAB deubiquitinating isopeptidases. Type III systems encode Cap6, homologous to eukaryotic Trip13 AAA⁺, and Cap7, homologous to eukaryotic HORMA domains. Some Type III systems additionally encode Cap8, a second divergent HORMA-like protein. Type IV systems encode proteins with predicted roles in converting and exchanging modified nucleobases: Cap9, a protein with a QueC enzymatic domain, and Cap10, a protein with a TGT enzymatic domain. Some Type IV systems additionally encode Cap11, a predicted N-glycosylase/DNA lyase. Cell-intrinsic immune pathways must connect pathogen sensing with a downstream immune response. Many classical pathways rely on direct protein–protein interactions between sensor proteins and adaptor proteins that activate transcription factors to induce antiviral gene programs (eukaryotes) or effector proteins that directly target invading viruses (prokaryotes). Animal cGLR-STING and bacterial CBASS pathways subvert this paradigm: Sensor cGLRs or CD-NTases communicate immune activation to downstream adaptor or effector proteins using nucleotide second messenger signals. Checked boxes indicate the evolutionary advantages to this form of immune signaling in either eukaryotes or prokaryotes, discussed throughout this review. Abbreviations: 2TM, two transmembrane helices-containing effector; AAA⁺, ATPases Associated with diverse cellular Activities; C5, chromosome 5; C6, chromosome 6; Cap, CD-NTase-associated protein; CBASS, cyclic nucleotide-based antiphage signaling system; CD-NTase, cGAS/DncV-like nucleotidyltransferase; CdnC/D/G/I, CD-NTase in Clade C/D/G/I; cGAS, cyclic GMP-AMP synthase; cGLR, cGAS-like receptor; hcGAS, human cGAS; HORMA, Hop1p, Rev7p, and Mad2 protein; hSTING, human STING; IFN, interferon; IRF3, interferon regulatory factor 3; NF- κ B, nuclear factor κ -light-chain-enhancer of activated B cells; NucC, nuclease, CD-NTase associated; QueC, Queuosine biosynthesis protein; STING, stimulator of interferon genes; TGT, tRNA-guanine transglycosylase; Trip13, thyroid hormone receptor interactor 13.

evolutionary history from CBASS to cGLR-STING immune systems provides a new foundation to explain broadly conserved mechanisms of antiviral immunity.

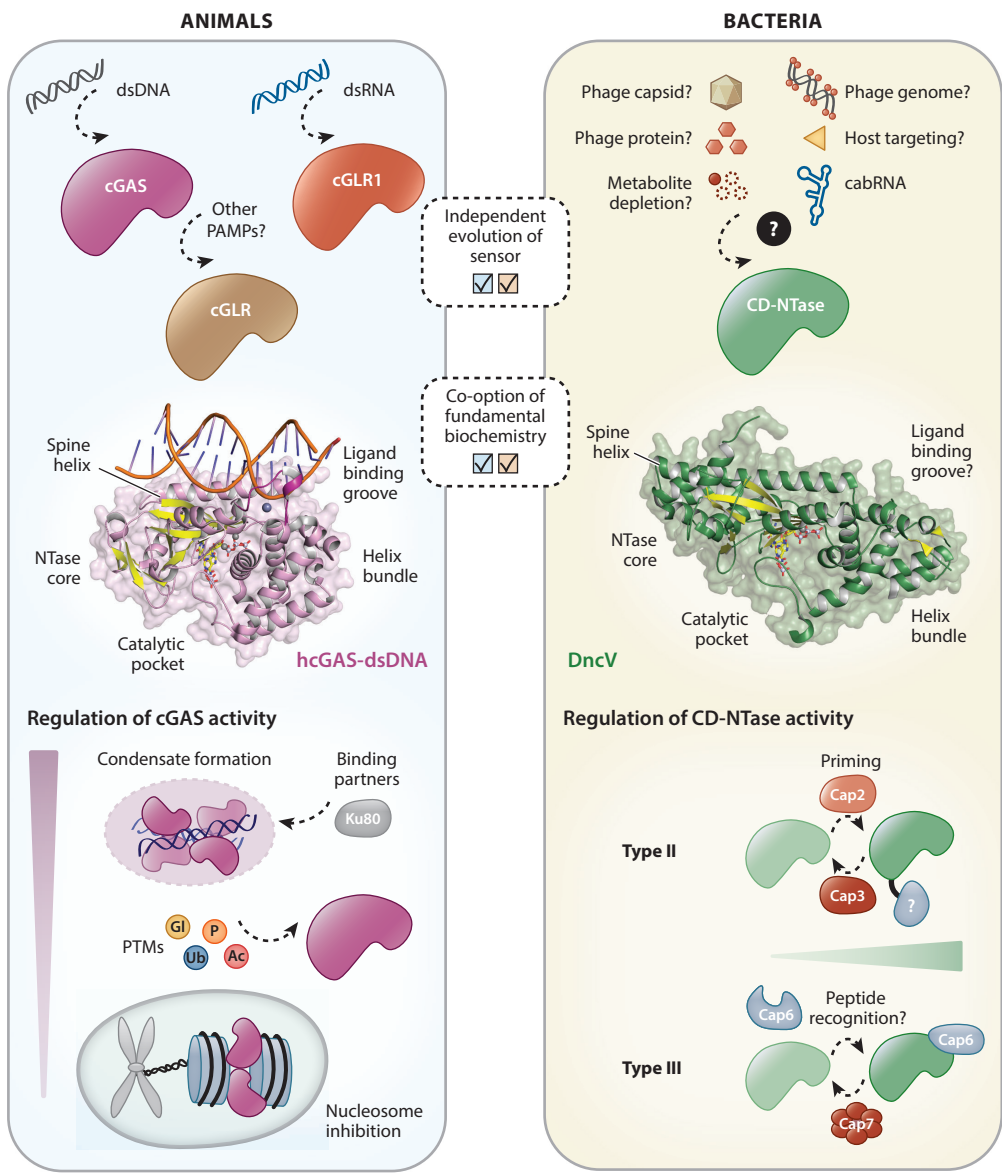
2. PATHOGEN SENSING AND REGULATION OF cGLR AND CD-NTase ACTIVITY

2.1. Detection of Infection by cGLRs and CD-NTases

Animal cGLRs and bacterial CD-NTases function as molecular sensors of pathogen infection. Animal cGLRs directly detect DNA and RNA as pathogen-associated molecular patterns (PAMPs) (Figure 2). In animal cells, DNA and RNA are important PAMPs, as compartmentalization and modification of nucleic acid are used to regulate cellular function and define self from nonself. Cytosolic dsDNA is a potent danger signal to metazoan cells and can result from self (micronuclei formation and rupture, loss of mitochondrial integrity, cellular stress events) and nonself (viral or bacterial genomic material) sources (8, 30). Human and vertebrate cGAS proteins function as direct sensors of cytosolic dsDNA to play important roles in antitumor immunity and the response to a broad spectrum of viral and bacterial pathogens (2, 31, 32). The discovery of cGLRs as a diverse family of pattern recognition receptors in animals revealed that divergent proteins in this family also function as dsRNA sensors (9, 10). dsRNA is a common PAMP recognized by

insect cGLRs including *Drosophila* cGLR1 and a suite of highly divergent invertebrate cGLRs in *Cnidaria*, hydra, and thrips (9–11). To date the only characterized PAMPs recognized by cGLRs are dsDNA and dsRNA, but many other PAMPs including lipopolysaccharide and other microbial cell wall components commonly stimulate innate immunity in animals, and it is predicted that the cGLR protein family likely recognizes other, as-yet-unidentified molecular cues of infection.

In bacteria, CD-NTases are activated to synthesize nucleotide second messengers upon phage infection, but for most enzymes the stimulating viral cue is currently unknown (**Figure 2**). The more than 5,000 identified bacterial CD-NTases are exceptionally diverse at the amino acid level (<10% sequence identity) and form at least eight major enzyme clades (Clades A–H) (12). Extreme



(Caption appears on following page)

Figure 2 (Figure appears on preceding page)

Pathogen sensing and regulation of cGLR and CD-NTase activity. In animals, cGAS and cGLRs directly detect molecular signatures of pathogen infection, including dsDNA and dsRNA. In bacteria, the molecular mechanism by which CD-NTases detect phage infection is currently unclear. Recent evidence supports that a CD-NTase enzyme from Clade E, CdnE, directly detects the phage-derived cabRNA. (*Top inset box*) Pathogen sensing by cGLRs and CD-NTases evolves independently of downstream signaling components. Crystal structures of the human cGAS–DNA complex [PDB: 6CTA (44)] and the CD-NTase DncV [PDBs: 4TY0/4U0M (45, 46)] reveal close structural homology between animal cGLR enzymes and bacterial CD-NTases. A spine helix bridges an N-terminal nucleotidyltransferase core and a C-terminal helix bundle, enclosing a catalytic pocket that supports nucleotide second messenger synthesis. In cGAS, a Zn-ribbon motif inserts into the ligand binding groove to control dsDNA recognition. β -Sheets are highlighted in yellow; the Zn-ribbon is highlighted in magenta. (*Bottom inset box*) cGLRs and CD-NTase enzymes are ancestrally related to DNA-pol β family enzymes that carry out fundamental nucleotidyltransferase biochemistry, including DNA replication and RNA repair. Recognition of long dsDNAs by cGAS dimers controls enzyme oligomerization. The cGAS unstructured N-terminal tail (not shown) further drives the formation of molecular condensates, which promote robust enzymatic activity. cGAS binding partners including Ku proteins further promote cGAS–DNA binding and condensate formation. Inhibitory PTMs downregulate cGAS enzymatic activity, and direct interactions with nucleosomes lock cGAS into a catalytically inactive conformation. In some CBASS systems, accessory Cap proteins regulate CD-NTase enzymatic activity. In Type II systems, Cap2 proteins prime nucleotide second messenger synthesis by conjugating the CD-NTase to an unknown protein substrate; Cap3 reverses this conjugation to downregulate activity. In Type III systems, peptide recognition by Cap6 promotes Cap6–CD-NTase complex formation to activate nucleotide second messenger synthesis; Cap7 disassembles the Cap6–CD-NTase complex. Abbreviations: Ac, acetyl-group; cabRNA, CBASS-activating bacteriophage RNA; Cap, CD-NTase-associated protein; CBASS, cyclic nucleotide-based antiphage signaling system; CD-NTase, cGAS/DncV-like nucleotidyltransferase; CdnE, CD-NTase in Clade E; cGAS, cyclic GMP-AMP synthase; cGLR, cGAS-like receptor; DncV, dinucleotide-cyclase in *Vibrio*; ds, double-stranded; Gl, glycan; hcGAS, human cGAS; P, phosphate; PAMP, pathogen-associated molecular pattern; PDB, Protein Data Bank; PTM, post-translational modification; Ub, ubiquitin; Zn, zinc.

divergence of bacterial CD-NTases suggests that distinct mechanisms likely exist for how CBASS immune systems sense phage infection. The prominent role of DNA sensing in cGAS-dependent animal immunity has led to speculation that bacterial CD-NTases may respond to viral genomic DNA. Although bacteria lack nuclei and genomic compartmentalization, unique base modifications in viral nucleic acids or replication intermediates associated with viral genome replication may provide an opportunity for accurate sensing of foreign DNA (33). However, while activation of a CD-NTase enzyme from *Escherichia coli* has been reported to be enhanced by DNA (18), there is currently no direct evidence in support of phage genome sensing by CD-NTases. Other nucleic acid species produced during viral replication may serve as molecular cues of infection. It was recently reported that second messenger synthesis by a *Staphylococcus schleiferi* CD-NTase from Clade E is directly activated by a unique, highly structured viral RNA produced during staphylococcal phage infection, termed the cabRNA (CBASS-activating bacteriophage RNA) (34). This discovery highlights a conserved role for nucleic acid immunity in eukaryotic and prokaryotic domains of life, and nucleic acid sensing by CD-NTases remains an important area of investigation. Interestingly, selection for phages resistant to CBASS immunity and related bacterial immune systems demonstrated that mutations in viral capsid proteins enable evasion of antiphage defense, suggesting that detection of phage capsids may trigger CBASS activation (26, 34, 35). Direct sensing of phage structural proteins in bacterial Avs-mediated immunity and by CapRel toxin-antitoxin operons suggests that conserved viral structures can be broadly recognized as PAMPs by bacterial defense systems (36, 37). However, staphylococcal capsid mutant phages that resist CBASS immunity still produce the CD-NTase-activating RNA, suggesting that capsid mutations mediate immune escape downstream of second messenger production (34). Finally, an important distinction between cGLRs and CD-NTases is that many bacterial CD-NTases are constitutively active in vitro when purified in isolation and in the absence of bound ligands (12). Consistent with this basal in vitro activity, it has been proposed that some CBASS systems function through a guard mechanism in which the depletion of a specific host factor during infection relieves enzymatic inhibition and stimulates nucleotide second messenger synthesis (14, 38). Understanding

how CD-NTases detect phage infection remains one of the most important open questions in nucleotide second messenger immune signaling.

2.2. Structural Features of cGLRs and CD-NTases

Animal cGLRs and bacterial CD-NTases share a conserved protein architecture in which an N-terminal nucleotidyltransferase core and C-terminal α -helix bundle are bridged by a spine helix to enclose a caged catalytic pocket that supports nucleotide second messenger synthesis (39) (**Figure 2**). A conserved catalytic triad ([DE]h [DE]h [X50–90] h[DE]h, where h is a hydrophobic amino acid) within this pocket controls nucleotidyltransferase activity. cGLR and CD-NTase enzymes are members of the DNA-pol β superfamily that includes ancestral proteins involved in DNA replication and RNA repair, demonstrating how the fundamental biochemical function of nucleotidyltransferase activity has been co-opted for an immune signaling role.

cGAS and cGLRs directly bind PAMPs through a conserved ligand binding groove along the spine helix (4, 9, 40–44). Bacterial CD-NTases contain a similar putative ligand binding groove, but it is unclear if this feature is responsible for recognition of a phage-associated cue (12, 45, 46). In cGAS and dsRNA-sensing cGLRs, including *Drosophila* cGLR1, positively charged residues along this groove are responsible for electrostatic interactions with nucleic acids. The cGAS ligand binding groove is further defined by a zinc coordinating motif insertion (Zn-ribbon), which is required for recognition of dsDNA ligands (4, 40, 41, 43, 44). The Zn-ribbon insertion is exclusive to vertebrate cGAS enzymes, explaining the ability of these proteins to function as dsDNA sensors. Intriguingly, invertebrate cGLRs from the species *Crassostrea gigas* and *Crassostrea virginica* that are closely related to vertebrate cGAS also function as dsDNA sensors. These proteins lack a Zn-ribbon and therefore may sense DNA through a distinct mechanism (11). Diversification of the ligand binding groove throughout metazoan cGLRs likely explains how divergent enzymes detect different molecular patterns (11).

The mechanism of mammalian cGAS activation is well characterized including defined interaction sites controlling recognition of dsDNA, formation of a minimally active 2:2 cGAS–DNA complex, and oligomerization into higher-order states critical for robust 2'3'-cGAMP synthesis (4, 40–44, 47–49). cGAS–DNA interactions are further controlled by an unstructured N-terminal domain that contributes to the formation of liquid condensates to promote cGAS enzymatic activity (48, 50). The molecular mechanism of ligand recognition and activation for other metazoan cGLRs and bacterial CD-NTases is currently uncharacterized. While the global architecture of this protein family is highly conserved, key structural differences between divergent cGLRs and CD-NTases likely control unique aspects of ligand recognition and enzymatic activation.

Unstructured N-terminal extensions are common among vertebrate cGAS proteins, and a subset of other metazoan cGLRs also contain disordered N or C termini. While most cGLRs are encoded as individual enzymatic domains, a select number of cGLRs are appended to second cGLR proteins, or other domains frequently associated with immunity including tetratricopeptide- and ankyrin-repeat modules (11). Ancillary domain fusions and disordered extensions are rare in prokaryotic CD-NTases. A notable exception are the ~30% of CD-NTases in Type I CBASS operons found fused to a domain of undefined function named AGS-C, which is predicted to have homology to Ig-fold proteins (15, 51, 52). While such N- and C-terminally appended domains in cGLRs and CD-NTases have no defined function, it is possible they contribute to the molecular detection of viral infection and the regulation of nucleotide second messenger synthesis.

2.3. Regulation of Enzymatic Activity

Regulation of cGAS enzymatic activation is critical to ensure correct sensing of cytosolic DNA in animal cells. Inappropriate activation of cGAS in the nucleus is prevented by a direct interaction

between cGAS and nucleosomal DNA where binding to the histone 2A-histone 2B acidic patch locks cGAS in a catalytically inactive state (53–58) (**Figure 2**). cGAS activity in the nucleus is further restrained by inhibitory interactions with the nuclear protein Baf (59). In the cell cytosol, a major form of cGAS regulation is the exonuclease TREX1 (three prime repair exonuclease 1) that degrades free cytosolic DNA and can also directly interact with cGAS–DNA condensates to limit immune activation (50, 60, 61). Activation of mammalian cGAS in the cytosol is further regulated by inhibitory post-translational modifications including ubiquitination, sumoylation, acetylation, glutamylation, and phosphorylation by Aurora kinase B during mitotic breakdown of the nuclear envelope (62–66). Numerous cellular and viral binding partners, including the putative DNA cosensors Ku80/Ku70 and ZCCHC3, have been implicated in regulating cGAS activity and appear to function through modulating the ability of cGAS to efficiently recognize DNA and form protein–DNA condensates (48, 67–72). These binding partners can be a target of viral immune antagonism; for example, the orthopoxvirus proteins C4 and C16 disrupt Ku–DNA interactions to inactivate cytosolic DNA sensing (73–75). Host regulation and viral antagonism of other animal cGLR signaling pathways remain to be characterized.

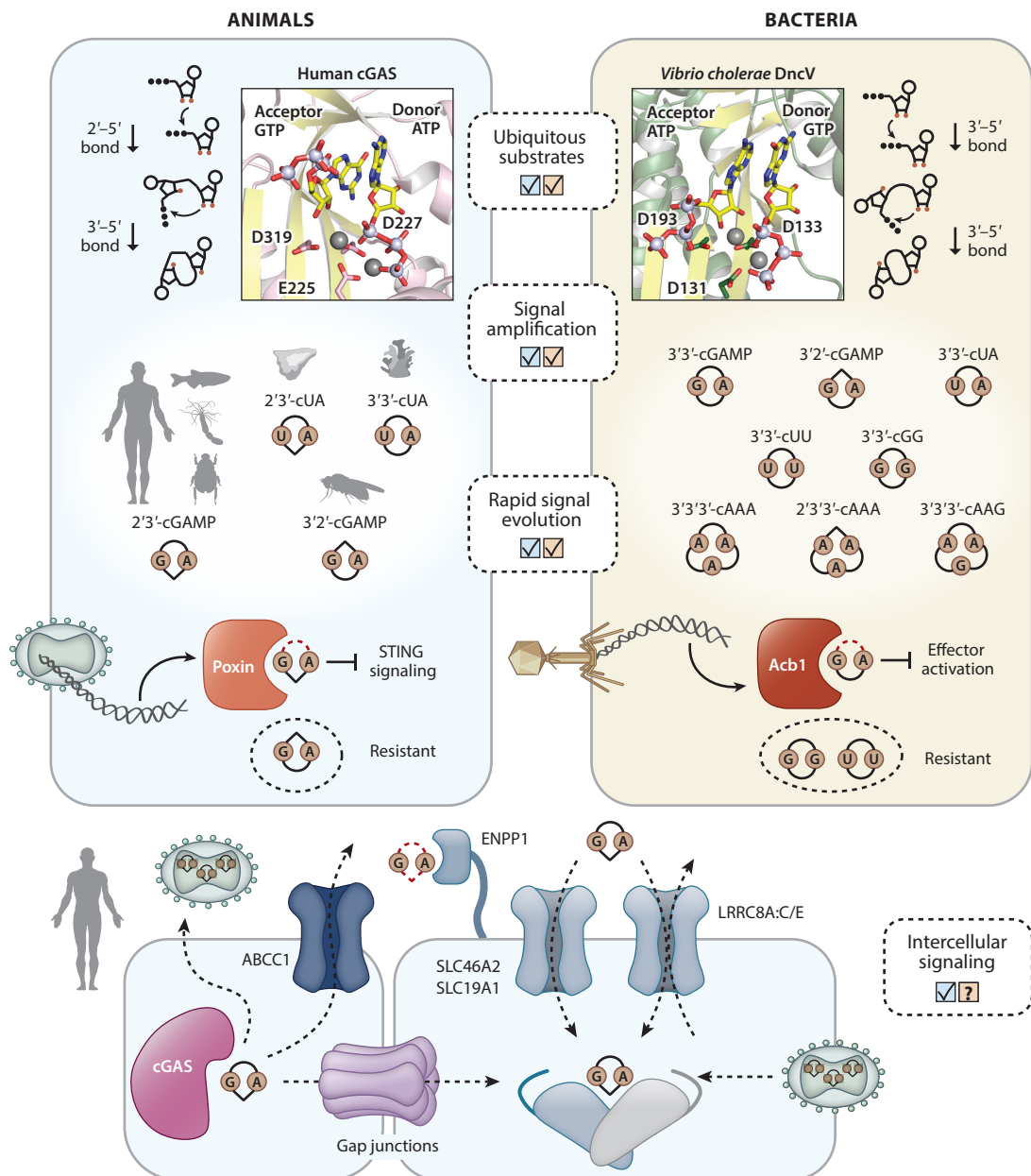
Growing evidence suggests that bacterial CD-NTase activation is regulated by auxiliary Cap proteins encoded within CBASS antiphage defense operons (**Figure 2**). Recently, multiple groups reported that Type II CBASS operons prime CD-NTase enzymatic activity using a reaction that shares remarkable similarity with animal ubiquitin transfer reactions (76–78). During CBASS antiphage defense, Cap2, an accessory protein with structural homology to eukaryotic E1 ubiquitin-activating and E2 ubiquitin-conjugating enzymes, covalently links the CD-NTase C terminus to an unknown substrate in a reaction that promotes a greater than 50-fold upregulation in nucleotide second messenger synthesis in response to phage infection. The Type II CBASS protein Cap3, an accessory protein with structural homology to eukaryotic JAB isopeptidase domains found in deubiquitinating enzymes, cleaves this linkage and downregulates CD-NTase enzymatic activity (76–78). Likewise, Cap6 and Cap7 proteins in Type III CBASS operons exhibit structural homology with eukaryotic TRIP13 (thyroid hormone receptor interactor 13) and HORMA (HOP1p, Rev7p, and MAD2) proteins and control a peptide recognition event that regulates CD-NTase activation during antiphage defense (18). Finally, operon transcription and CD-NTase expression can be regulated by WYL domain-containing transcription factors that tune overall gene expression (79). It remains an open question whether phages have evolved mechanisms to specifically target CBASS signaling at the step of CD-NTase activation.

3. NUCLEOTIDE SECOND MESSENGERS IN cGLR AND CBASS IMMUNITY

3.1. Mechanism of Nucleotide Second Messenger Synthesis

A defining feature of cGLR-STING and CBASS is the utilization of a nucleotide second messenger to link pathogen detection with downstream antiviral immunity. Following PAMP recognition, animal cGLRs and bacterial CD-NTases use a single catalytic site to catalyze a conserved multi-step reaction resulting in the production of a cyclic dinucleotide or cyclic trinucleotide nucleotide signal (4, 9, 12, 41, 44, 45) (**Figure 3**). Enzymatic activity is controlled by a conserved triad of aspartate and glutamate residues that coordinate either magnesium or manganese catalytic ions. In each reaction, the 5' α -phosphate of a nucleotide donor is linked to the 2' or 3' hydroxyl of a nucleotide acceptor (forming a 2'–5' or 3'–5' phosphodiester bond, respectively). For cyclic dinucleotide signals such as 2'3'-cGAMP and 3'3'-cGAMP, the resulting linear intermediate reorients within the catalytic pocket with donor and acceptor positions reversed for formation of the second bond and cyclization of the final product (4, 12, 45). Cyclic trinucleotides are formed through

an iterative reaction process, but it remains unclear how the resulting linear intermediates are reoriented prior to final cyclization (80, 81). cGLR and CD-NTase nucleotide product identity is controlled primarily by key residues within the active site lid region that form hydrogen-bond interactions with donor and acceptor nucleobases (12, 45, 80). Variability at these positions is an important driver of nucleotide second messenger product diversity and enables cGLR-STING and CBASS operons to rapidly evolve signal specificity to maintain antiviral immunity. Additional specificity determinants include a network of minor contacts that contribute to nucleotide



(Caption appears on following page)

Figure 3 (Figure appears on preceding page)

Nucleotide second messengers in cGLR and CBASS immunity. cGAS, cGLR, and CD-NTase enzymes synthesize nucleotide second messenger signals in a conserved multi-step reaction. (*Top inset box*) cGLR and CD-NTase immune signals are synthesized from common ribonucleotide substrates. A catalytic triad of aspartate (D) and glutamate (E) residues coordinate either magnesium or manganese catalytic ions to sequentially synthesize 2'-5' or 3'-5' phosphodiester linkages between nucleotide substrates. (*Left inset*) The human cGAS active site coordinating ATP and GTP substrates to synthesize the first pppG[2'-5']pA bond. (*Right inset*) The DncV active site coordinating substrates to synthesize the first pppA[3'-5']pG bond. (*Upper middle inset box*) The multi-turnover synthesis of nucleotide second messengers dramatically amplifies immune signaling. Confirmed animal cGLR signals are shown with corresponding source organisms: 2'/3'-cGAMP is synthesized in diverse animals including humans, vertebrate fish, beetles, and *Nematostella*; 2'/3'-cUA is synthesized in *Crassostrea gigas* and *Crassostrea virginica*; 3'/3'-cUA is synthesized in *Stylophora pistillata*; and 3'/2'-cGAMP is synthesized in *Drosophila*. Confirmed bacterial CD-NTase signals are shown, including diverse cyclic di- and trinucleotides. (*Lower middle inset box*) cGLR and CD-NTase signals rapidly evolve, and are likely to control divergent immune pathways and evade viral antagonism. Both animal viruses and bacteriophages encode nucleases that cleave cGLR and CD-NTase nucleotide second messengers to antagonize immune signaling. In animals, poxins efficiently cleaves 2'/3'-cGAMP but not 3'/2'-cGAMP. In bacteria, Acb1 efficiently cleaves nucleotide second messengers that incorporate an adenosine base but not 3'/3'-cGG or 3'/3'-cUU. In animals, the cGAS product 2'/3'-cGAMP plays an important role as an intercellular immune signal. 2'/3'-cGAMP is exported and imported through several channels, is incorporated into viral particles, and transits to neighboring cells through gap junctions. The host-encoded nuclease ENPP1 cleaves 2'/3'-cGAMP to regulate the available extracellular pool of this immune signal. (*Bottom inset box*) Intercellular signaling is an evolutionary advantage of nucleotide second messenger immune signals demonstrated in animals but not in bacteria. Abbreviations: ABCC1, ATP-binding Cassette subfamily C member 1; Acb1, anti-CBASS protein 1; cAAA, cyclic AMP-AMP-AMP; cAAG, cyclic AMP-AMP-GMP; CBASS, cyclic nucleotide-based antiphage signaling system; CD-NTase, cGAS/DncV-like nucleotidyltransferase; cGAMP, cyclic GMP-AMP; cGAS, cyclic GMP-AMP synthase; cGG, cyclic GMP-GMP; cGLR, cGAS-like receptor; cUA, cyclic UMP-AMP; cUU, cyclic UMP-UMP; DncV, dinucleotide-cyclase in *Vibrio*; ENPP1, ectonucleotide pyrophosphatase phosphodiesterase 1; LRRC8, Leucine-Rich-Repeat containing protein family 8; poxins, poxvirus immune nuclease; SLC19A1, Solute Carrier family 19 member 1; SLC46A2, Solute Carrier family 46 member 2; STING, stimulator of interferon genes.

selection and specific nucleotide rotation that directs either 2'-5' or 3'-5' phosphodiester linkage formation (see the sidebar titled 2'-5' Phosphodiester Linkages) (80).

Nucleotide second messenger synthesis provides distinct advantages within cGLR-STING and CBASS antiviral signaling pathways. First, ribonucleotide substrates are ubiquitous throughout the cell, and thus cGLRs and CD-NTases can rapidly synthesize large quantities of second messenger molecules. Second, cGLR and CD-NTase activation dramatically amplifies initial signaling. For example, activation of a few enzymes of cGAS bound to DNA causes multi-turnover production with more than 750 molecules of 2'/3'-cGAMP signal synthesized for each active enzyme (4). Likewise, in response to phage infection, CD-NTases synthesize micromolar levels of nucleotide second messenger signals to rapidly activate cell death effector Cap proteins (13, 17).

2'-5' PHOSPHODIESTER LINKAGES

While 3'-5' phosphodiester bonds are the common linkage in cellular RNA and DNA, 2'-5' phosphodiester bonds are rare in biology. 3'-5' and 2'-5' phosphodiester linkage specificity is critical to the ability of cGLR and CD-NTase nucleotide second messengers to interact with receptors with high affinity and control downstream immune signaling (7, 17, 86). 2'-5' phosphodiester linkages also play a key role in differentiating nucleotide immune signaling pathways and preventing cross talk with other endogenous host cell processes. Notably, a role for 2'-5' linkages in immune signaling was first identified in the linear 2'-5' oligoadenylate product synthesized by mammalian OAS1 (oligoadenylate synthase 1) (164, 165). OAS1 is a double-stranded RNA sensor that signals through the downstream receptor RNase L to block viral replication (29). Although initial structures of human cGAS revealed structural similarity with OAS1, cGAS and animal cGLRs share significantly greater structural homology with bacterial CD-NTases, demonstrating a closer evolutionary relationship between bacterial CBASS immunity and animal cGLR-STING signaling than with OAS1 RNA sensing (9).

Additionally, nucleotide second messenger synthesis in animals results in an elevated immune response that spreads as cGLR-produced nucleotide signals are transferred between gap junctions of neighboring cells and imported into immune cells (82–85). No extracellular roles are known for CD-NTase nucleotide signals in bacteria. Finally, nucleotide second messenger synthesis removes the need for direct protein–protein interactions and enables sensor cGLR and CD-NTase proteins to evolve independently from downstream immune adaptor and cell death effector proteins. Most animals encode multiple cGLR proteins (commonly 2–4 cGLRs, more than 40 cGLRs in some select species), and some bacterial CBASS operons encode two unique CD-NTases, suggesting pathogen sensing can be enhanced by sensor protein duplication (11, 12). In animal cGLR-STING immunity, duplication of adaptor genes additionally allows cGLR nucleotide second messengers to signal through multiple STING receptors and potentially distinct downstream pathways (11). Likewise, in bacterial CBASS, nucleotide signal production enables Cap proteins to be rapidly exchanged between operons to diversify CBASS systems and expand the repertoire of effectors used to inhibit phage replication (15, 17). Together these unique properties of nucleotide-based immune signaling help explain evolutionary selection of these pathways across divergent kingdoms of life.

3.2. Diversity of Nucleotide Second Messengers in Animals and Bacteria

cGLRs and CD-NTases synthesize a remarkable array of distinct nucleotide second messengers (Figure 3). 2'3'-cGAMP is the major product of vertebrate cGAS enzymes and is thought to be the most common nucleotide second messenger controlling cGLR-STING immunity in animals (5–7, 11, 86, 87). Biochemical screening of hundreds of cGLRs encoded throughout the metazoan kingdom demonstrates that both DNA- and RNA-sensing enzymes from diverse species synthesize 2'3'-cGAMP (11). Although 2'3'-cGAMP was initially assumed to be the exclusive nucleotide second messenger in metazoan pathways, it was recently discovered that *Drosophila* cGLR1 synthesizes a distinct chemical isomer of cGAMP, 3'2'-cGAMP (cG[3'–5']pA[2'–5']p) (9). cGLR1 synthesizes 3'2'-cGAMP using the opposite nucleobase reaction order compared to cGAS, selecting for the reverse donor and acceptor substrates in the synthesis of the first phosphodiester bond. Animal cGLR-produced nucleotide signals are now known to include pyrimidine substrates and distinct combinations of 3'–5' and 2'–5' phosphodiester linkages. For example, a cGLR from *C. virginica* synthesizes 2'3' cyclic UMP-AMP (2'3'-cUA) in response to the ligand dsDNA while a cGLR from *Stylophora pistillata* synthesizes 3'3'-cUA in response to the ligand dsRNA (11). Analysis of these cGLR enzymes underscores that a diversity of nucleotide second messenger signals likely remain to be discovered in animal biology.

Bacterial CBASS systems signal through an exceptionally wide array of distinct nucleotide second messengers, with 12 confirmed signals identified and more than 180 possible signals predicted (38). First discovered in *Vibrio cholerae*, the most common CD-NTase nucleotide signal is the product 3'3'-cGAMP (45, 88). However, bacterial CD-NTase enzymes can utilize all four nucleobases, A, C, G, and U, and can incorporate both 3'–5' and 2'–5' linkages (12, 17). CD-NTase nucleotide second messengers additionally include cyclic trinucleotide signals, with several CD-NTase enzymes, particularly in Clades C and D, synthesizing cyclic trinucleotides 3'3'3'-c-AMP-AMP-GMP (3'3'3'-cAAG) and 3'3'3'-c-AMP-AMP-AMP (3'3'3'-cAAA) (12, 89). Cyclic trinucleotides are also synthesized in Type III CRISPR immune systems, and CBASS and CRISPR operons share evolutionarily related effector proteins that respond to these nucleotide signals (17, 90). The mammalian cyclic nucleotide receptor RECON (reductase controlling NF-κB) can sense bacterial 3'3'3'-cAAG and 3'3'3'-cAAA (12, 91), but endogenous cyclic trinucleotide products have thus far not been identified in metazoan immunity. CBASS operons typically do not

signal through the common bacterial nucleotide second messengers 3′3′-c-di-AMP and 3′3′-c-di-GMP. Notable exceptions are some Clade E CD-NTase enzymes that synthesize 3′3′-c-di-GMP to activate STING-like Cap effectors to induce NAD⁺ (nicotinamide adenine dinucleotide) depletion and cell death (14). STING-containing CBASS operons occur exclusively in bacteria that are devoid of common endogenous 3′3′-c-di-GMP signaling enzymes, highlighting the necessity for CD-NTase nucleotide signals to be distinct from normal host responses in order to establish functional antiphage defense. Some nucleotide second messenger products, including 3′2′-cGAMP and 3′3′-cUA, are produced by both bacterial CD-NTases and metazoan cGLRs, and it is likely that new discoveries will further expand the mechanistic similarity between CBASS and cGLR-STING signaling (9, 11, 34, 51, 92).

3.3. Role of Nucleotide Messengers as Immunotransmitters

A unique aspect of nucleotide second messengers is the ability to control both intracellular and intercellular immune signaling (**Figure 3**). As charged small molecules, nucleotide second messengers cannot passively permeate cellular membranes. In animals, the cGAS product 2′3′-cGAMP can be transferred between gap junctions formed between closely linked cells (82). Additionally, 2′3′-cGAMP is actively exported by the channel ABCC1 (ATP-binding Cassette subfamily C member 1) and is released upon cell death to create an extracellular pool of nucleotide signal (93). Multiple channels including SLC19A1 (Solute Carrier family 19 member 1) and SLC46A2 (Solute Carrier family 46 member 2) family channels as well as LRRC8A:C/E (Leucine-Rich-Repeat containing protein family 8 A:C/E) heterodimers import 2′3′-cGAMP and are particularly important for the role of cGAS and 2′3′-cGAMP signaling in antitumor immunity (83–85, 94). Different 2′3′-cGAMP import channels have varied cell type-specific expression profiles and exhibit distinct biochemical properties that influence the ability of cells to respond to extracellular nucleotide signaling (85). The level of extracellular 2′3′-cGAMP is regulated by the enzyme ENPP1 (ectonucleotide pyrophosphatase phosphodiesterase 1) (95, 96). ENPP1 exists as a dimer tethered to the extracellular membrane or as a secreted protein abundant in plasma and robustly cleaves both 2′–5′ and 3′–5′ bonds of extracellular 2′3′-cGAMP, rendering GMP and AMP molecules that cannot activate STING (96). Notably, 2′3′-cGAMP can also confer longer-range cell-cell immunity as a passenger in viral particles. Upon cGAS activation in virally infected cells, 2′3′-cGAMP is packaged in progeny virions and delivered to the cytoplasm of subsequently infected cells where it directly activates STING to initiate antiviral immunity (97, 98).

To date, intercellular signaling roles have not yet been defined for other cGLR or CBASS nucleotide second messengers in animals or bacteria. However, studies in *Drosophila* support uptake of extracellular cyclic dinucleotide signals (99), and given that many of the characterized mammalian transport channels are broadly conserved in the metazoan lineage, it is likely that other cGLR products function as immunotransmitters much like 2′3′-cGAMP. Despite living as single-cell organisms, bacteria frequently grow in clonal patches and communicate with other colony members using extracellular chemical communication systems (e.g., quorum sensing). However, given that CBASS nucleotide second messengers control effector-induced cell death, it is unclear that CD-NTase products would function as advantageous intercellular signals.

3.4. Viral Antagonism of Nucleotide Signals

An evolutionary vulnerability of nucleotide second messenger signaling is susceptibility of cGLR and CD-NTase nucleotide signals to viral inhibition (100, 101) (**Figure 3**). In animals, mammalian poxviruses, insect baculoviruses, select insect flaviviruses, and lepidopteran host species encode a 2′3′-cGAMP-specific nuclease named poxin (poxvirus immune nuclease) (100, 102, 103). Poxin

recognizes 2'3'-cGAMP with nucleobase and phosphodiester linkage specificity and cleaves the 3'-5' bond through metal-independent hydrolysis to inactivate host signaling. 2'3'-cGAMP cleavage by poxin is critical for robust poxvirus replication in certain mammalian hosts, underscoring both the importance of cGAS-STING signaling for *in vivo* antiviral defense and the significant evolutionary pressure for viruses to overcome this pathway (100, 102, 104). In a striking evolutionary parallel, diverse phages encode the nuclease Acb1 (anti-CBASS protein 1) that cleaves CD-NTase nucleotide signals and inhibits downstream CBASS immune activation in bacteria (101). Although Acb1 and poxin share no sequence or structural homology, Acb1 similarly cleaves 3'-5' linkages through metal-independent hydrolysis to inactivate nucleotide second messenger immune signaling (101). Notably, Acb1 degrades a broad range of CD-NTase products, including 3'3'-cGAMP, 3'3'-cUA, and 3'3'3'-cAAA, but does not cleave common host nucleotide signals such as 3'3'-c-di-GMP. Some cGLR and CD-NTase products are resistant to viral nuclease cleavage, including 3'2'-cGAMP and 3'3'-c-di-UMP, suggesting that host signaling pathways have correspondingly evolved to escape viral targeting of immune signaling (9, 101).

In addition to enzymatic degradation, an alternative viral evasion strategy is nucleotide second messenger sequestration. Analysis of the bacterial cyclic nucleotide defense system Thoeris demonstrated that phages can encode sponge inhibitors that tightly bind and sequester nucleotide immune signals to terminate signaling and prevent downstream receptor activation (28). Two recent reports discovered the phage protein Acb2 as a nucleotide sponge that forms a hexameric complex that binds 3'3'-cGAMP and prevents CBASS immune activation (35, 78). No eukaryotic viruses are yet known to encode nucleotide second messenger sponges, but it is likely that this evasion mechanism also exists in animal viruses to inhibit cGLR-STING signaling.

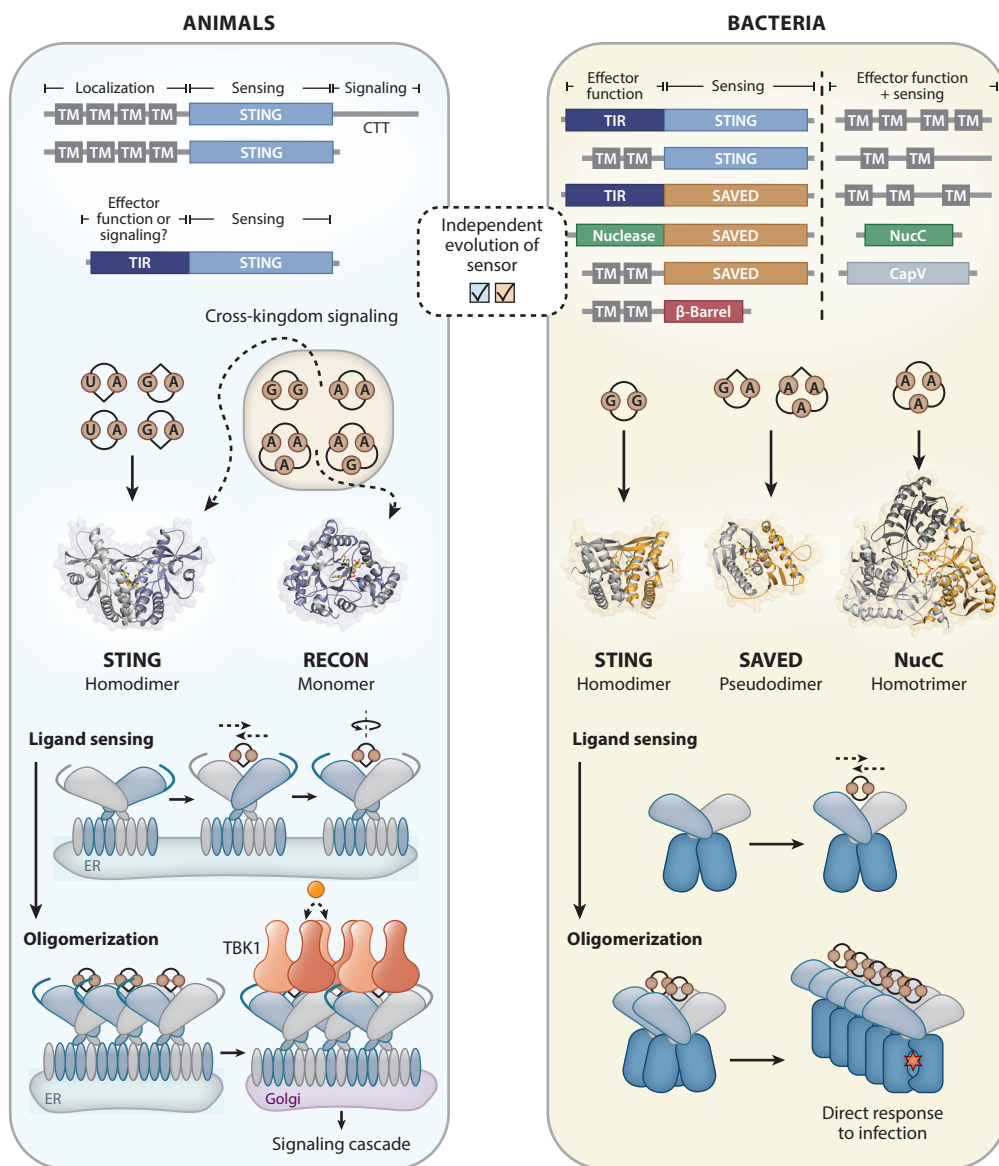
4. DETECTION OF NUCLEOTIDE IMMUNE SIGNALS BY DOWNSTREAM RECEPTORS

4.1. Functional Domains in Animal and Bacterial Nucleotide Second Messenger Receptors

Nucleotide second messengers synthesized by cGLRs and CD-NTases directly bind and activate receptors to control downstream immunity through signaling cascades or direct effector functions (**Figure 4**). In metazoans, STING is the major receptor of endogenously synthesized cGLR nucleotide signals. Animal STING proteins form obligate homodimers and contain N-terminal transmembrane (TM) domains that localize the receptor to the ER (endoplasmic reticulum), a central ligand binding domain that binds nucleotide second messengers, and an unstructured C-terminal tail (CTT) that controls the recruitment and activation of downstream signaling kinases (1, 3, 105, 106). Invertebrate STING proteins do not contain a CTT, and thus it is currently unclear how these receptors control immune signaling (87, 107, 108). The STING cyclic dinucleotide binding domain evolved in bacteria as a CBASS receptor, and some bacterial STING proteins are appended to a TIR (Toll-interleukin-1 receptor) domain that cleaves the essential metabolite NAD⁺ and causes cell death (14). Intriguingly, several invertebrate STING proteins also contain an N-terminal TIR domain, but the immune mechanism of animal TIR-STING proteins remains an open question (14).

In bacterial CBASS immunity, most Cap protein nucleotide second messenger receptors are composed of discrete sensor domains that directly bind nucleotide second messengers and effector domains that target essential cellular machinery to control antiphage defense (17). The best-characterized sensor domains include bacterial STING domains (Cap12 and Cap13) (14, 109), a domain named SAVED (synthase-associated and fused to various effector domains) that is a fusion of two CARF (CRISPR-associated Rossmann fold) domains found in CRISPR immunity

(Cap4 and Cap5) (17, 92, 110), and a small β -barrel nucleotide binding domain (Cap15) (51). Effector domains include nucleases, TM domains, phospholipases, TIR NADases, and more rarely peptidases and phosphatases (15, 17, 52). Many CBASS effector domains are common to other antiphage defense systems, highlighting conserved strategies of antiviral immunity and illustrating how genetic exchange between mobile elements in prokaryotes fuels the host-pathogen arms race (17, 90). For a subset of receptors, such as NucC (nuclease, CD-NTase associated), a single domain carries out both sensor and effector functions (89). Other rarer CBASS receptors encode distinct domains that have yet to be fully characterized. Importantly, the modularity of sensor and effector domains in CBASS receptors allows for the rapid evolution of defense mechanisms



(Caption appears on following page)

Figure 4 (Figure appears on preceding page)

Detection of nucleotide immune signals by downstream receptors. Specific domains in animal STING and bacterial CBASS receptor proteins control nucleotide second messenger sensing and downstream signaling or effector functions. In animal STING proteins the central ligand binding domain controls nucleotide second messenger sensing and N-terminal TM domains control localization to the ER membrane. In vertebrate STING proteins an unstructured CTT recruits kinases to control downstream immune signaling. TIR domains in some invertebrate STING proteins have an uncharacterized immune mechanism. In bacteria, CBASS Cap receptors are composed of modular configurations of domains that sense nucleotide second messengers and domains with cell death effector functions. The receptor proteins shown in the figure above correspond with the following Cap names: TIR-STING, Cap12; TM-STING, Cap13; “nuclease,” REase-SAVED, Cap4; HNH-SAVED, Cap5; TM-SAVED, Cap14; TM- β -barrel, Cap15. 4 TM protein refers to S4TM, 2 TM protein refers to SLATT4, and 3 TM protein refers to SLATT6. Diverse protein scaffolds in animals and bacteria specifically recognize cGMP and CBASS nucleotide second messenger signals. (Inset box) Nucleotide second messenger receptors are able to evolve independently of upstream pathogen-sensing enzymes. Crystal structures of human STING in complex with 2'3'-cGAMP [PDB: 4KSY (7)] and mouse RECON in complex with cAAG [PDB: 6M7K (12)] demonstrate how nucleotide second messenger-sensing domains in animals recognize either endogenously produced immune signals or bacterially produced cross-kingdom signals. Structurally characterized nucleotide second messenger receptors in bacteria include STING in complex with 3'3'-c-di-GMP [PDB: 6WT4 (14)], SAVED in complex with 2'3'3'-cAAA [PDB: 6VM6 (17)] [see also SAVED in complex with 3'2'-cGAMP (92)], and NucC in complex with 3'3'3'-cAAA [PDB: 6Q1H (89)]. Receptor protein oligomerization upon the sensing of nucleotide second messenger ligands is a conserved feature of animal cGMP-STING and bacterial CBASS signaling. In animal STING and bacterial TIR-STING (Cap12), conformational changes upon direct binding to nucleotide second messenger ligands induce head-to-head interactions between STING dimers and drive the formation of filaments. Vertebrate animal STING filament formation leads to the clustering of the CTT and the recruitment of the immune signaling kinase TBK1. Bacterial TIR-STING filament formation promotes cross-dimer TIR contacts to activate TIR NADase activity. Abbreviations: cAAA, cyclic AMP-AMP-AMP; cAAG, cyclic AMP-AMP-GMP; Cap, CD-Nase-associated protein; CBASS, cyclic nucleotide-based antiphage signaling system; cGMP, cyclic GMP-AMP; cGMP, cGAS-like receptor; CTT, C-terminal tail; ER, endoplasmic reticulum; HNH, nuclease with histidine (H) and asparagine (N) catalytic residues; NADase, nicotinamide adenine dinucleotide hydrolase; NucC, nuclease, CD-Nase associated; PDB, Protein Data Bank; REase, restriction endonuclease; RECON, reductase controlling NF- κ B; SAVED, synthase-associated and fused to various effector domains; SLATT, synthase and LOG-Smf/DprA-associating two TM; STING, stimulator of interferon genes; TBK1, TANK-binding kinase 1; TIR, Toll-interleukin-1 receptor; TM, transmembrane.

in response to a specific nucleotide signal. This aspect of CBASS immunity creates evolutionary plasticity that allows for immune responses to be separately fine-tuned at the nodes of pathogen sensing, immune signal synthesis, signal reception, and pathogen targeting.

4.2. Molecular Mechanism of Nucleotide Second Messenger Recognition

Diverse protein domains have evolved to specifically recognize the vast diversity of nucleotide second messengers synthesized in cGMP-STING and CBASS immunity (**Figure 4**). In animal immunity, the major nucleotide second messenger receptor STING forms a homodimeric, V-shaped complex with a deep central pocket for cyclic dinucleotide recognition (7, 9, 11, 14, 86, 87, 111, 112). Upon high-affinity recognition of endogenous cGMP-produced nucleotide signals, the STING β -strand lid closes over the top of the bound ligand and each STING monomer rotates inward to create a compact configuration that seals the nucleotide second messenger binding pocket (7, 9, 86). A network of protein-ligand contacts controls selective recognition of cyclic dinucleotides including stacking interactions between aromatic residues and each nucleobase, and polar contacts at the base of the pocket that read out phosphodiester linkage specificity (7, 9, 11, 14, 86, 87, 111, 112). Substitutions of specific polar residues at the base of the ligand binding pocket explain the specificity of animal STING proteins for distinct endogenous cGMP signals (9, 11), and variations in conserved arginine residues within human STING are known to alter nucleotide second messenger affinity (7, 86). Most animals (88%) encode a single STING protein; however, some animal species have an expanded repertoire of cGMP-STING pathways including the coral *S. pistillata*, which encodes 42 cGMPs and 7 STINGs (11). Structural remodeling of the ligand binding pocket in *S. pistillata* STING proteins controls recognition of distinct nucleotide

signals produced by different *S. pistillata* cGLRs, demonstrating how STING specificity can be leveraged to coordinate complex immune signaling networks.

In addition to detection of endogenous immune signals, animal STING proteins can also more weakly recognize bacterial cyclic dinucleotide ligands including 3′/3′-c-di-GMP, 3′/3′-c-di-AMP, and the CBASS signal 3′/3′-cGAMP (5, 6, 105). In humans, recognition of exogenously derived bacterial cyclic dinucleotides is important for pathogen detection (113–115) and sensing of gut microbiota (116), and potentially in antitumor immunity (117). In another example of cross-kingdom signaling, the nucleotide second messenger receptor RECON is an oxidoreductase in mice that responds to the bacterial cyclic dinucleotide 3′/3′-c-di-AMP and controls NF-κB signaling (91). RECON additionally recognizes CBASS signals including 3′/3′-cUA and the cyclic trinucleotide signals 3′/3′/3′-cAAA and 3′/3′/3′-cAAG (12), but thus far the receptor is not known to respond to endogenous cGLR signaling products. RECON functions as a monomer, binding nucleotide second messengers in the same pockets that control adenine and nicotinamide recognition during normal oxidoreductase catalytic function (91). Several RECON homologs exist in human cells, but these proteins have no defined function, suggesting possible yet-undiscovered roles in cGLR signaling and animal innate immunity.

STING proteins are functional receptors in CBASS immunity, demonstrating that STING cyclic dinucleotide recognition originated in bacteria (14). Bacterial and animal STING proteins exhibit striking structural homology and share a conserved mechanism of signal recognition within the homodimeric V-shaped ligand binding pocket. All characterized bacterial STING proteins selectively recognize 3′/3′-c-di-GMP, suggesting that the STING homodimer configuration may have originally evolved to bind twofold symmetric nucleotide ligands (14, 109, 118). STING receptors in CBASS are rare, accounting for nucleotide second messenger recognition in less than 5% of CBASS operons (14). The most common CBASS receptors contain a nucleotide second messenger binding domain named SAVED (present in ~30% of all operons) (17, 52). SAVED domains recognize nucleotide ligands as a monomeric unit, and divergent domains have been shown to respond to asymmetric ligands including 2′/3′/3′-cAAA and 3′/2′-cGAMP (17, 92). Crystal structures of the CBASS receptor protein Cap4 demonstrated that the SAVED domain is a pseudodimer fusion of two ancestral CARF proteins that function in Type III CRISPR systems as sensors of cA₄ and cA₆ nucleotide second messengers (17). Prototypical CARF domains unite as a homodimer to sense symmetric nucleotide ligands, suggesting that duplication and fusion into the monomeric form SAVED enabled the diversification of the nucleotide ligand binding surfaces to allow for the recognition of asymmetric ligands. SAVED domains have also been identified in Type III CRISPR operons, suggesting that a conserved mechanism of nucleotide ligand recognition allows for evolutionary exchange between these two antiphage defense systems (90). In addition to STING and SAVED, some sensor domains appear to be unique to CBASS, including the Cap15 β-barrel protein that functions as one of the smallest protein units able to specifically recognize cyclic dinucleotides (51). For some CBASS receptors a single domain functions as both the sensor and effector. For example, the receptor NucC binds the ligand 3′/3′/3′-cAAA through a symmetric allosteric pocket created by loops at the base of the nuclease homotrimer, demonstrating how the surface of a functional domain can be co-opted for nucleotide messenger sensing and how receptor oligomers can be harnessed to sense ligands with corresponding symmetry (89). Several major families of CBASS receptors remain to be characterized at the structural level including the phospholipase CapV that responds to 3′/3′-cGAMP (119) and several classes of enigmatic TM segment-containing receptors including SLATT (synthase and LOG-Smf/DprA-associating two TM) and S4TM (4 TM) proteins (51, 52). Defining the molecular basis of receptor activation in CBASS immunity will continue to provide an important foundation to understand conserved principles of nucleotide second messenger recognition in immune defense.

4.3. Nucleotide Second Messenger Sensing Controls Receptor Oligomerization and Activation

In both animal cGLR-STING and bacterial CBASS nucleotide second messenger signaling, the major mechanism of receptor activation is protein oligomerization and filament formation (**Figure 4**). Recent landmark studies defined the structural basis of oligomerization and activation of full-length animal STING proteins (106, 120). In cGAS-STING signaling, 2'3'-cGAMP binds deep within the central STING ligand binding pocket to induce the compact closed conformation of the homodimer. Following 2'3'-cGAMP recognition and conformational change, a transition in the connector helix between the STING ligand binding domain and the N-terminal TM helices results in a 180° rotation. These conformational changes promote head-to-head interactions between neighboring STING dimers and drive the formation of higher-order filaments along the ER membrane surface (106). In addition to 2'3'-cGAMP-induced oligomerization, trafficking of STING from the ER to Golgi via COPII (Coat Protein Complex II) vesicles is critical to initiate downstream immune signaling (1, 121–123). STING filament formation results in clustering of the unstructured STING C-terminal tails that recruit dimers of the kinase TBK1 and induce TBK1 *trans*-autophosphorylation and downstream immune signaling (120, 124).

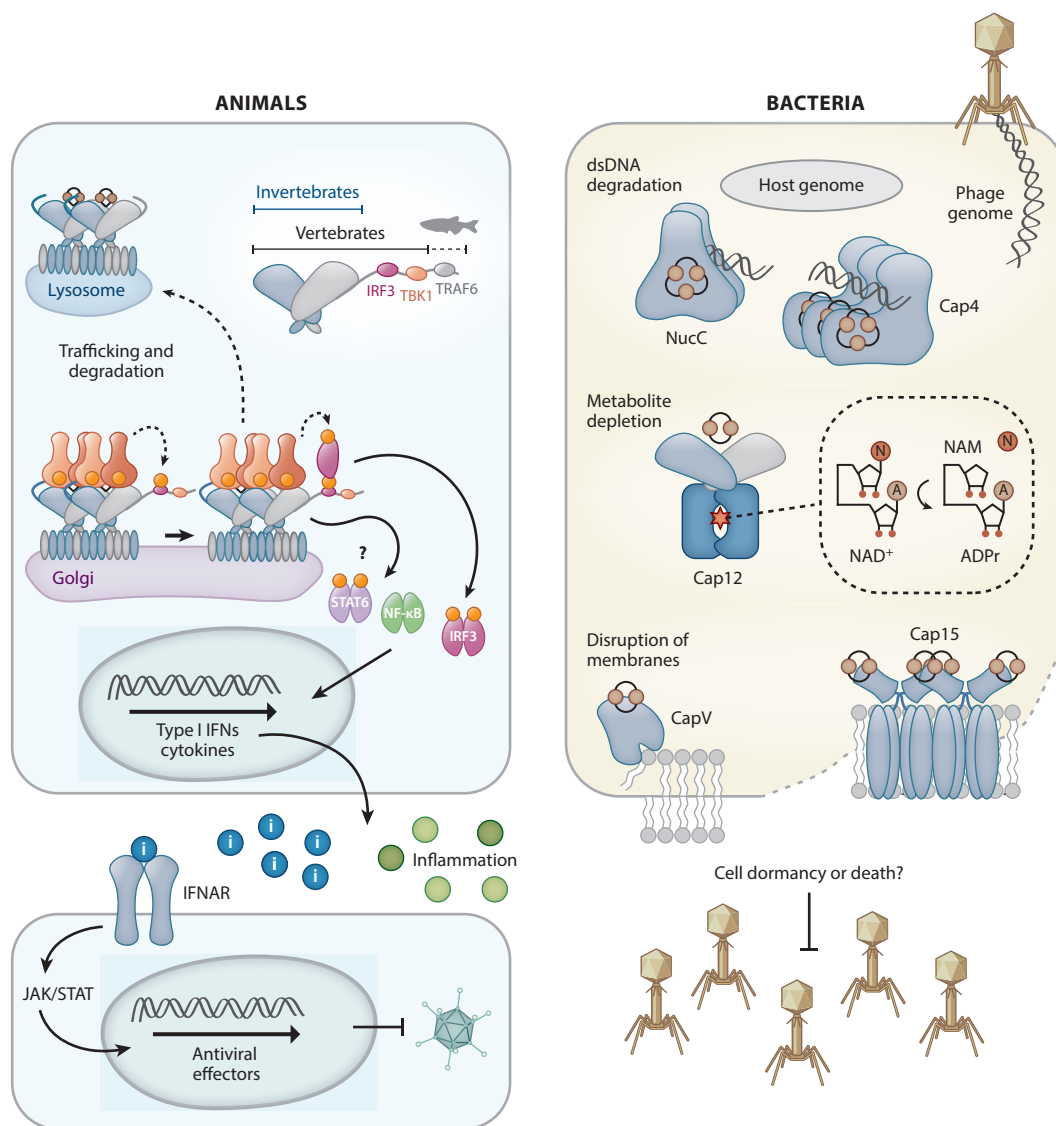
The requirement of receptor oligomerization to control downstream immunity is conserved in bacterial CBASS (17). Structural analysis of bacterial Cap12 TIR-STING filament formation reveals striking conservation with the mechanism of metazoan STING activation (109). In response to the CD-NTase signal 3'3'-c-di-GMP, the bacterial STING domain adopts a compact closed conformation that exposes oligomerization interfaces and enables head-to-head STING filament formation. Bacterial TIR-STING oligomerization is further stabilized by cross-dimer contacts that occur at surfaces conserved in both bacteria and mammalian STING proteins. In CBASS immunity, TIR-STING filament formation promotes cross-dimer contacts in the TIR domain that activate the NADase catalytic site to trigger rapid NAD⁺ hydrolysis and inhibition of phage replication (109). Analysis of the CBASS effectors Cap4 [REase (restriction endonuclease)-SAVED] and Cap12 (TIR-SAVED) demonstrate that filament formation is also required for activation of SAVED domain-containing effectors (17, 110). Adjacent SAVED domains sandwich the bound nucleotide signals to create a stacking interaction that induces receptor filament formation and activation of appended cell death effector domains (92, 110). CBASS TM effectors including Cap15 also require oligomerization for activation and bacterial inner membrane disruption (51). Although filaments have not been observed for the receptor NucC, recognition of 3'3'3'-cAAA by this receptor controls hexamer formation that is required for effector nuclease activity (89). The large structural diversity of receptor domains that function through nucleotide second messenger-induced filament formation highlights the key role of receptor oligomerization in rapid activation of CBASS antiphage defense.

5. MECHANISMS OF ANTIVIRAL DEFENSE IN cGLR-STING AND CBASS IMMUNITY

5.1. STING Signal Transduction and Antiviral Immunity

In animals, cGLR-STING signaling controls downstream immunity through the induction of antiviral gene programs (**Figure 5**). Mammalian STING signal transduction primarily relies on the kinase TBK1, which is recruited following STING filament formation and Golgi translocation by a conserved PLPLRT/SD motif in the STING C-terminal tail (120, 124). Upon binding to clustered STING C-terminal tails, TBK1 dimers activate through *trans*-autophosphorylation and proceed to phosphorylate an adjacent STING C-terminal tail motif PLxIS that then recruits the transcription factor IRF3 (120, 125, 126). Subsequent IRF3 phosphorylation by TBK1 licenses

IRF3 to dimerize and translocate to the nucleus, where it stimulates the transcription of type I IFNs. STING recognition of 2'3'-cGAMP also activates the transcription factors NF- κ B and STAT6 (signal transducer and activator of transcription 6) to control proinflammatory cytokine production, but the mechanisms of these signaling events remain poorly understood (8, 127, 128). Interestingly, while the major signaling output of mammalian STING is the induction of type I IFNs, ray-finned fish STING proteins selectively induce NF- κ B signaling through the addition of a noncanonical TRAF6 (TNF-receptor associated factor 6) motif on the C terminus of the CTT (107). Thus, in vertebrate species, the STING CTT functions as a modular signaling platform in which discrete motifs control the activation of downstream signaling responses (107). Following IRF3 and NF- κ B activation at the Golgi, STING traffics to the lysosome for degradation as a



(Caption appears on following page)

Figure 5 (Figure appears on preceding page)

Mechanisms of antiviral defense in cGAS and CBASS immunity. Animal cGLR-STING signaling protects cells from infection through the induction of antiviral gene programs. The vertebrate STING CTT functions as a signaling platform, with distinct motifs to recruit the kinase TBK1 and the transcription factor IRF3. The STING CTT in ray-finned fish further has a motif to recruit TRAF6. Invertebrate STING proteins do not have a CTT. TBK1 dimers are recruited to the CTT following STING 2'3'-cGAMP binding and activate through a *trans*-autophosphorylation event. TBK1 then phosphorylates the upstream IRF3 motif to license the recruitment of IRF3, which is in turn phosphorylated and activated by TBK1. STING nucleotide second messenger binding further activates the transcription factors STAT6 and NF- κ B through an unknown mechanism. Active STAT6, NF- κ B, and IRF3 dimers translocate to the nucleus and induce the transcription of cytokines and type I IFNs, resulting in inflammation, the induction of antiviral ISGs, and the stimulation of adaptive immunity. Following the activation of transcription factors at the Golgi membrane, STING traffics to the lysosome for degradation. Bacterial CBASS immunity relies on the degradation of essential cellular components by effector proteins. CBASS effectors include endonucleases, such as NucC and Cap4, that promiscuously degrade dsDNA; enzymes that degrade important cellular metabolites, such as Cap12 proteins that cleave NAD⁺; and proteins that disrupt inner membrane integrity either through TM domains, as does Cap15, or through phospholipase activity, as does CapV. The degradation of essential cellular components leads to cell death or dormancy, thereby limiting the production of progeny phage particles and halting viral spread throughout the bacterial population. Abbreviations: ADPr, adenine diphosphate ribose; Cap, CD-NTase-associated protein; CBASS, cyclic nucleotide-based antiphage signaling system; cGAMP, cyclic GMP-AMP; cGAS, cyclic GMP-AMP synthase; cGLR, cGAS-like receptor; CTT, C-terminal tail; ds, double-stranded; IFN, interferon; IFNAR, interferon (IFN- α/β) receptor; IRF3, interferon regulatory factor 3; ISG, interferon-stimulated gene; JAK, Janus kinase; NAD⁺, nicotinamide adenine dinucleotide; NAM, nicotinamide; NF- κ B, nuclear factor κ -light-chain-enhancer of activated B cells; NucC, nuclease, CD-NTase associated; STAT, signal transducer and activator of transcription; STAT6, signal transducer and activator of transcription 6; STING, stimulator of interferon genes; TBK1, TANK-binding kinase 1; TM, transmembrane; TRAF6, TNF-receptor associated factor 6.

mechanism to attenuate immune signaling in the absence of continuous nucleotide second messenger synthesis (121, 122). A number of post-translational modifications and partner proteins have been described to play a role in STING trafficking and signal transduction, reviewed here (129).

STING is conserved throughout animals as the receptor for cGLR-produced nucleotide second messengers. However, the STING CTT signaling platform and type I IFNs are exclusive to the vertebrate lineage. It has been proposed that in most primitive metazoan species the major signaling output of STING is NF- κ B signaling and induction of autophagy (87, 130–136), thereby suggesting that the structural requirements for activating this pathway are present in the core STING ligand binding domain. In *Drosophila* and *Nematostella*, NF- κ B has been demonstrated *in vivo* to control STING-mediated immune signaling, with recent studies in both organisms profiling the complete transcriptional response to STING activation (133, 135). STING-induced autophagy has also been described as a primordial function of cGLR-STING signaling common across all animals (134). In human cells, STING induces autophagy by increasing LC3 lipidation at the ER-Golgi intermediate compartment, and this signaling process has a functional role in clearance of DNA and pathogens from the cell cytosol (134). The motifs required for STING-induced autophagy map to an α -helix conserved at the outer face of the animal STING ligand binding domain that is not present in bacterial STING (14, 134).

STING-induced NF- κ B and IFN signaling is a key node of animal innate immunity that coordinates a potent response to pathogen detection. In vertebrates, inflammatory cytokine induction by NF- κ B creates localized inflammation to stimulate immune response pathways in neighboring cells and promote the recruitment of professional immune cells (137, 138). NF- κ B in more divergent animal species additionally controls the expression of direct antibacterial and antiviral effector proteins (133, 135). Vertebrate type I IFNs activate autocrine and paracrine signaling through IFNAR-JAK-STAT [interferon (IFN- α/β) receptor-Janus kinase-signal transducer and activator of transcription] signal transduction, resulting in the coordinated expression of hundreds of antiviral effectors termed ISGs (interferon-stimulated genes). ISGs play an essential role in combatting pathogen infection, working either alone or in concert to directly target viral

machinery, block viral replication, or disrupt host pathways that are required for the viral life cycle (139). Such strategies for protecting host cells from infection are evolutionarily ancient, and many ISGs may have originally evolved as part of bacterial antiphage defense systems (140, 141).

5.2. CBASS Effector-Mediated Antiphage Defense

CBASS effectors control downstream immunity by degrading essential cellular components to cause cell death or dormancy, thereby limiting the production of progeny phage particles and halting viral spread (13, 17, 18) (**Figure 5**). Broadly named abortive infection, induction of host cell death is a common strategy among bacterial defense systems to protect the surrounding clonal population from infection (19). Anti-phage defense systems are often encoded together in genomic defense islands and are spread through horizontal gene transfer between distantly related bacteria (142). In addition to defending against diverse viral threats, defense systems such as CBASS are therefore also selected for the ability to function in a wide range of divergent genetic backgrounds. As such, CBASS effectors often cause abortive infection by targeting universally conserved biomolecules such as nucleic acids, NAD^+ and other essential metabolites, and the inner cell membrane.

Endonucleases are common effectors in Type II, III, and IV CBASS systems, representing 30% of Type II effectors, 71% of Type III effectors, and 23% of Type IV effectors (15, 17, 89). CBASS effector nucleases include Cap4 (REase-SAVED), Cap5 {HNH [nuclease with histidine (H) and asparagine (N) catalytic residues]-SAVED}, NucC, and rarer Mrr-like or calcineurin-like proteins. CBASS Cap4 contains a divergent restriction endonuclease (REase) domain (previously annotated DUF4297) that cleaves dsDNA (17). Cap4 exists in an inactive monomeric state, and recognition of the cognate CD-NTase nucleotide second messenger signal by the Cap4 SAVED domain drives oligomerization and filament formation. Cap4 oligomerization results in REase domain clustering and enzymatic activation of promiscuous dsDNA cleavage activity that degrades the host chromosome and viral DNA to small ~ 17 -bp fragments (17). CBASS effector Cap5, which contains the HNH-nuclease domain, functions through a similar mechanism of action where SAVED-dependent nucleotide second messenger recognition drives enzyme oligomerization and DNase effector function (17, 92). The CBASS nuclease NucC is a constitutive homotrimer with each monomer unit containing an REase active site positioned on the outer edge of the oligomeric complex. Upon binding the CD-NTase signal 3'3'3'-cAAA, NucC assembles into an active hexamer that degrades dsDNA substrates to ~ 50 – 100 -bp fragments (89). Cap4, Cap5, and NucC nucleases each rely on nucleotide second messenger-dependent oligomerization to control activation of promiscuous enzymatic domains and cleave dsDNA targets regardless of circularization, methylation pattern, or source of origin (17, 89, 92). Notably, CBASS nuclease effectors are frequently encoded in other antiphage defense systems, with the Cap4 REase domain found in Lamassu and AVAST (17, 143, 144), and NucC found in Type III CRISPR systems (89, 145, 146).

Depletion of essential metabolites is another strategy through which CBASS effectors control antiphage abortive infection. TIR domains from plants, animals, and bacteria cleave NAD^+ to inhibit cellular redox reactions and block central metabolism (14, 147, 148). CBASS TIR effectors are fused to the nucleotide second messenger sensor domains STING (Cap12) and SAVED. Like CBASS nuclease effectors, TIR proteins exist as inactive monomers or dimers in the absence of nucleotide second messenger signaling and form active oligomerized filament complexes following nucleotide signal recognition (14, 109, 110). CBASS TIR effectors are exceptionally potent NADases *in vitro*, cleaving NAD^+ at more than 10 times the rate of plant and animal TIR proteins involved in immune defense and causing rapid growth arrest when activated in cells (14). NAD^+ cleavage is a widespread cell death mechanism in prokaryotic defense systems, with TIR domain

effectors also occurring in Pycsar, Thoeris, and SIR2 (silent information regulator 2)-based defense systems (26, 27, 149). Beyond NADases, CBASS effectors include nucleosidases that have been shown to degrade ATP (150). CBASS effectors are also predicted to include phosphatases and peptidases that may target other biomolecules required for cell viability (15, 52).

An especially diverse class of CBASS effectors is cell death proteins that disrupt the inner cell membrane (51, 119). TM domain-containing effectors (TM effectors) are some of the most common CD-NTase nucleotide second messenger receptors, making up nearly 40% of all CBASS effectors and appearing frequently in Type I (67%) and Type IV (81%) CBASS systems (15, 51). CBASS TM effectors exhibit diverse protein domain architectures including as TM segments fused to discrete nucleotide binding domains or small TM segment-only proteins with small intracellular loops. Structural and functional analysis of the CBASS effector Cap15 demonstrated that the protein contains two TM segments fused to a minimal β -barrel nucleotide second messenger binding domain (51). Cyclic dinucleotide recognition induces Cap15 oligomerization and a clustering event that causes inner membrane rupture and rapid cell death. The CBASS effector Cap14 contains TM segments fused to a SAVED domain and likely functions through similar oligomerization-dependent membrane disruption (51). Other CBASS TM effectors include SLATT4, SLATT6, and S4TM domain-containing proteins that consist of 2–4 TM segments (52). It is not known how these minimal TM effectors control nucleotide second messenger recognition or cause membrane disruption. A *Caulobacter crescentus* CheY-like protein involved in bacterial motility regulation recognizes 3'3'-c-di-GMP with a minimal, linear 19 amino acid motif (151), suggesting that CBASS SLATT and S4TM TM effectors may use intervening soluble protein loops to coordinate nucleotide second messenger binding. Recent analysis reveals that a TM effector in CRISPR immunity named Csx28 forms membrane pores (152), but no current evidence supports that CBASS TM effectors can form membrane-spanning pores. Instead, analysis of Cap14 and Cap15 suggests that inner membrane disruption occurs through raft-like clustering that results in large membrane lesions (51). CBASS effectors also target inner membrane integrity through enzymatic domains. CapV is a patatin-like phospholipase effector common in Type II CBASS operons that upon 3'3'-cGAMP recognition cleaves cell membrane phospholipids to control cell death and inhibit phage replication (13, 119). The targeting of cellular membrane integrity may be an especially efficient mechanism for inhibiting phage replication, as the chemical properties of inner membrane integrity are highly conserved across diverse bacteria and the precise timing of host cell lysis is critical for successful virion propagation.

6. EVOLUTION OF CBASS AND cGLR IMMUNE SYSTEMS ACROSS DOMAINS OF LIFE

6.1. CBASS Evolution in Prokaryotes

Prokaryotic immune systems, including CBASS operons, tend to cluster in dynamic genomic hotspots of antiviral immunity termed defense islands (153, 154) (Figure 6). While defense islands may be maintained in bacterial populations by the vertical genetic transfer of host chromosomes, they are also frequently encoded on mobile genetic elements that move through bacterial populations by diverse mechanisms of horizontal gene transfer (155, 156). In this dynamic ecosystem of genetic exchange, the evolutionary pressures of phage infection select for potent antiviral systems, drive diversification of defense island composition, and promote the spread of defense systems across divergent bacterial populations. As a result, CBASS systems are widespread in prokaryotic genomes, encoded alongside a diversity of companion defense systems, and control antiphage immunity through diverse mechanisms of nucleotide signal synthesis and Cap effector-induced cell death.

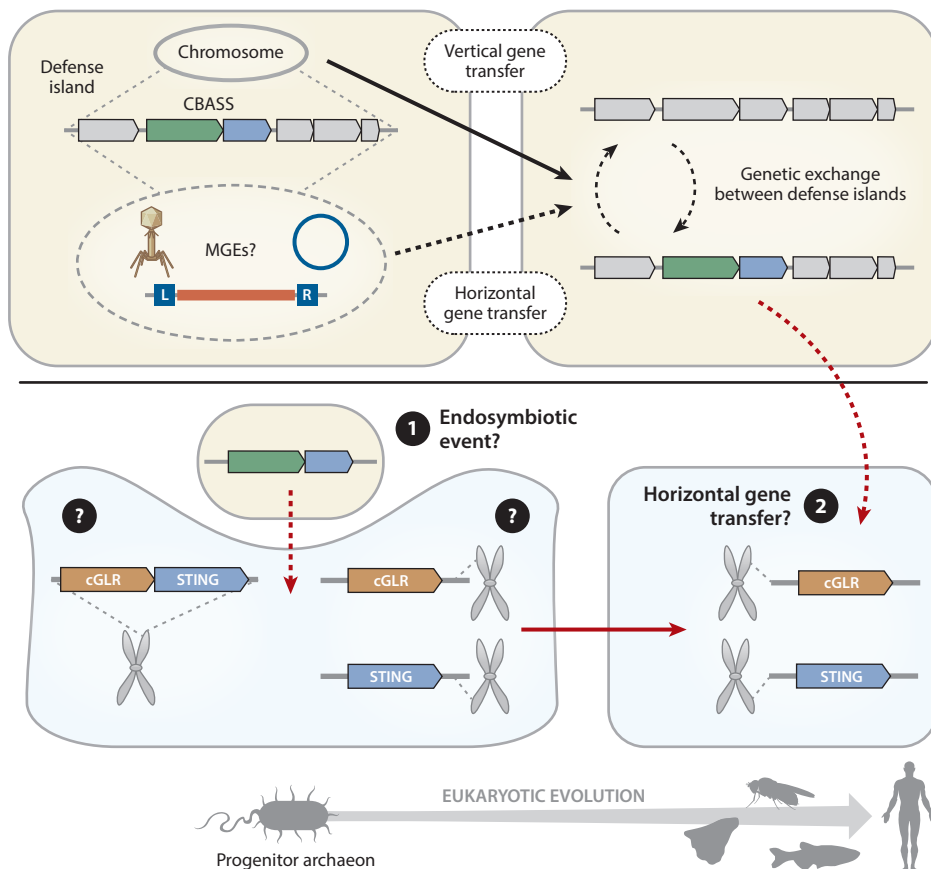


Figure 6

Evolution of CBASS and cGLR immune systems across domains of life. In bacteria, CBASS systems are encoded in defense islands on host chromosomes or MGEs such as prophage, transposons, and plasmids. Defense islands move through bacterial populations by vertical and horizontal gene transfer with a gene composition rapidly evolving through genetic exchange. Data in the field support two evolutionary scenarios for the acquisition of *cGLR* and *STING* genes by the metazoan lineage from ancestral CBASS systems: In scenario 1, *cGLR* and *STING* genes were acquired by a progenitor archaeon from a CBASS system in alpha-proteobacteria during endosymbiotic events that founded the eukaryotic lineage. As eukaryotes evolved, cGLRs and STING were retained by metazoans and lost in plants, fungi, and protists. In scenario 2, *cGLR* and *STING* genes were acquired by an ancestral metazoan through a rare event of horizontal gene transfer from bacteria. A further open evolutionary question is whether cGLRs and STING were acquired jointly, from a single CBASS operon, or individually, from distinct operons. cGLRs have evolved throughout animals to form a diverse family of pattern recognition receptors, and STING proteins have evolved to control different mechanisms of antiviral immunity. Abbreviations: CBASS, cyclic nucleotide-based antiphage signaling system; cGLR, cGAS-like receptor; MGE, mobile genetic element; STING, stimulator of interferon genes.

6.2. Evolutionary Acquisition of cGLRs and STING by Metazoans

Structural and functional conservation establishes CBASS antiphage defense systems as the ancestral origin of animal cGLR-STING antiviral pathways (12–14). *cGLR* and *STING* genes are encoded in most animal species (11, 87, 108) and in choanoflagellates (136) but have not been identified in other eukaryotes. The mechanism of evolutionary acquisition of progenitor

CD-NTase enzymes and STING nucleotide second messenger receptors by the metazoan lineage remains a major open question. Evidence in the field supports two potential evolutionary models (Figure 6):

1. *cGLR* and *STING* genes were members of a primordial arsenal of eukaryotic innate immune genes. This reservoir was seeded during a series of endosymbiotic events between archaea and alpha-proteobacteria that likely originated the eukaryotic lineage (157–159). As eukaryotes evolved, this repertoire of immune genes then diversified throughout different lineages, with the loss of select systems and the evolution of retained systems through domain shuffling, gene duplication, and mutation. In this model, cGLRs and STING were present in an early eukaryote and specifically lost in plant, fungi, and protist lineages.
2. *cGLR* and *STING* genes were acquired through rare cross-kingdom horizontal gene transfer events between bacteria and an early metazoan progenitor (160, 161). In support of this model, *cGLR* and *STING* genes have not been identified in plants, fungi, or the majority of unicellular eukaryotes (108). Choanoflagellates, the closest living relatives of animals, are the most divergent eukaryotes encoding STING, supporting that they may have been near the original node of acquisition of CBASS components (136).

A further open question in the evolutionary acquisition of cGLR-STING by metazoans is whether each component gene was acquired together from a single CBASS operon or separately through individual events. In one evolutionary scenario, acquisition of a complete CD-NTase-STING operon from bacteria created a functional signaling pathway that controlled cell death as an immune response to pathogen infection. Genome evolution in the metazoan lineage could then have separated *cGLR* and *STING* genes to distinct chromosomes and enabled gene duplication and diversification events that created the current diversity of cGLR-STING signaling pathways. STING is exceptionally rare in bacteria, occurring in less than 3% of CBASS operons (14). It is therefore unclear why an evolutionary acquisition of CBASS into metazoans would include STING and not more common Cap effectors. In an alternative evolutionary scenario, STING was first acquired from bacteria and functioned alone as a receptor to sense common bacterial cyclic dinucleotide PAMPs including 3'3'-c-di-GMP and 3'3'-c-di-AMP. In support of a STING-first evolutionary event, bacterial STING receptors are the only CBASS effectors that respond to 3'3'-c-di-GMP (14), and STING-dependent sensing of foreign 3'3'-c-di-GMP and 3'3'-c-di-AMP is known to be a critical component of the animal immune response to bacterial infection (113–115). Once cells encoded a functional STING receptor to sense nucleotide second messenger signals, a strong selective advantage may have then existed to further acquire CD-NTase proteins and establish endogenous cGLR-STING signaling pathways. In either scenario, the diverse repertoire of cGLRs encoded across animals may have evolved from a signal CD-NTase progenitor or multiple distinct CD-NTases that seeded different animal cGLR lineages.

A final evolutionary puzzle is how 2'3'-cGAMP emerged as the major nucleotide second messenger in animal cGLR-STING immunity. 2'3'-cGAMP is synthesized by diverse cGLRs throughout the animal kingdom (11), but no bacterial CD-NTases have been identified that produce 2'3'-cGAMP. Although the most common CBASS signal is 3'3'-cGAMP, CD-NTases can signal through 2'-5'-linked products, suggesting that 2'3'-cGAMP may occur in bacteria as a rare signal in antiphage defense (17, 92). Early acquisition of a 2'3'-cGAMP-synthesizing CD-NTase into eukaryotes may have created the foundation for the major role of this signal in animal immunity. Alternatively, 2'3'-cGAMP may have independently evolved in eukaryotes as a mechanism to discriminate endogenous signaling pathways from bacterial nucleotide second messenger detection.

6.3. cGLR-STING Evolution in Metazoans

Animal cGLRs are extremely diverse at the amino acid level, with most enzymes exhibiting less than 30% sequence identity. Metazoans typically encode 1–5 cGLRs; however, select phyla exhibit extreme genetic radiation, with species such as the oyster *C. virginica* encoding more than 200 cGLRs (11). Conversely, cGLRs have been lost in certain animal phyla, including nematodes and platyhelminthes. Intriguingly, the cGLR protein scaffold also appears to have been co-opted in animals for nonimmune functions. Widely conserved Mab21-like proteins are close structural homologs of cGLRs that lack a complete enzymatic site and appear to play a role in tissue development (162, 163). Most animals encode a single STING protein (11, 108). Unlike CBASS TIR-STING and TM-STING effectors that control cell death, animal STING proteins control downstream immunity via autophagy and the induction of antiviral gene programs. Throughout the course of metazoan evolution, STING acquired unique structural features to control animal innate immunity, including the C-terminal tail present in vertebrate STING proteins (14). The ancestral immune mechanism of STING and the protein features that control immune activation in invertebrate STING proteins remain an important area of investigation.

7. CONCLUDING REMARKS

The use of nucleotide second messenger signals to link pathogen detection with immune activation is a conserved mechanism of antiviral immunity shared between the prokaryotic and eukaryotic kingdoms of life. Bacterial CBASS and animal cGLR-STING immune systems illustrate the unique evolutionary advantages of using nucleotide second messenger immune signals, including the co-option of fundamental nucleotidyltransferase biochemistry, the ubiquity of ribonucleotide substrates, rapid signal evolution, signal amplification, and the independent evolution of cGLR and CD-NTase proteins from downstream immune adaptor or cell death effector proteins. The discovery of the evolutionary origins of cGAS-STING signaling reveals a direct connection between animal immunity and bacterial antiphage defense and provides a framework to explain the rules that govern host-virus interactions across all domains of life.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENTS

We thank members of the Kranzusch lab for their helpful comments and discussion. P.J.K. acknowledges grant support from the Pew Biomedical Scholars program, the Burroughs Wellcome Fund PATH program, The Mathers Foundation, The Mark Foundation for Cancer Research, the Cancer Research Institute, the Parker Institute for Cancer Immunotherapy, and the National Institutes of Health (1DP2GM146250-01). K.M.S. is supported as an NCI F99 Graduate Fellow (NIH 1F99CA274660-01).

LITERATURE CITED

1. Ishikawa H, Barber GN. 2008. STING is an endoplasmic reticulum adaptor that facilitates innate immune signalling. *Nature* 455:674–78
2. Sun L, Wu J, Du F, Chen X, Chen ZJ. 2013. Cyclic GMP-AMP synthase is a cytosolic DNA sensor that activates the type I interferon pathway. *Science* 339:786–91
3. Zhong B, Yang Y, Li S, Wang YY, Li Y, et al. 2008. The adaptor protein MITA links virus-sensing receptors to IRF3 transcription factor activation. *Immunity* 29:538–50

4. Gao P, Ascano M, Wu Y, Barchet W, Gaffney BL, et al. 2013. Cyclic [G(2',5')pA(3',5')p] is the metazoan second messenger produced by DNA-activated cyclic GMP-AMP synthase. *Cell* 153:1094–107
5. Ablasser A, Goldeck M, Cavlar T, Deimling T, Witte G, et al. 2013. cGAS produces a 2'-5'-linked cyclic dinucleotide second messenger that activates STING. *Nature* 498:380–84
6. Diner EJ, Burdette DL, Wilson SC, Monroe KM, Kellenberger CA, et al. 2013. The innate immune DNA sensor cGAS produces a noncanonical cyclic dinucleotide that activates human STING. *Cell Rep.* 3:1355–61
7. Zhang X, Shi H, Wu J, Zhang X, Sun L, et al. 2013. Cyclic GMP-AMP containing mixed phosphodiester linkages is an endogenous high-affinity ligand for STING. *Mol. Cell* 51:226–35
8. Ablasser A, Chen ZJ. 2019. cGAS in action: expanding roles in immunity and inflammation. *Science* 363:eaat8657
9. Slavik KM, Morehouse BR, Ragucci AE, Zhou W, Ai X, et al. 2021. cGAS-like receptors sense RNA and control 3'2'-cGAMP signalling in *Drosophila*. *Nature* 597:109–13
10. Holleufer A, Winther KG, Gad HH, Ai X, Chen Y, et al. 2021. Two cGAS-like receptors induce antiviral immunity in *Drosophila*. *Nature* 597:114–18
11. Li Y, Slavik KM, Morehouse BR, de Oliveira Mann CC, Mears K, et al. 2023. cGLRs are a diverse family of pattern recognition receptors in animal innate immunity. *bioRxiv* 2023.02.22.529553. <https://doi.org/10.1101/2023.02.22.529553>
12. Whiteley AT, Eaglesham JB, de Oliveira Mann CC, Morehouse BR, Lowey B, et al. 2019. Bacterial cGAS-like enzymes synthesize diverse nucleotide signals. *Nature* 567:194–99
13. Cohen D, Melamed S, Millman A, Shulman G, Oppenheimer-Shaanan Y, et al. 2019. Cyclic GMP-AMP signalling protects bacteria against viral infection. *Nature* 574:691–95
14. Morehouse BR, Govande AA, Millman A, Keszei AFA, Lowey B, et al. 2020. STING cyclic dinucleotide sensing originated in bacteria. *Nature* 586:429–33
15. Millman A, Melamed S, Amitai G, Sorek R. 2020. Diversity and classification of cyclic-oligonucleotide-based anti-phage signalling systems. *Nat. Microbiol.* 5:1608–15
16. Tesson F, Herve A, Mordret E, Touchon M, d'Humieres C, et al. 2022. Systematic and quantitative view of the antiviral arsenal of prokaryotes. *Nat. Commun.* 13:2561
17. Lowey B, Whiteley AT, Keszei AFA, Morehouse BR, Mathews IT, et al. 2020. CBASS immunity uses CARF-related effectors to sense 3'-5'- and 2'-5'-linked cyclic oligonucleotide signals and protect bacteria from phage infection. *Cell* 182:38–49
18. Ye Q, Lau RK, Mathews IT, Birkholz EA, Watrous JD, et al. 2020. HORMA domain proteins and a Trip13-like ATPase regulate bacterial cGAS-like enzymes to mediate bacteriophage immunity. *Mol. Cell* 77:709–22
19. Lopatina A, Tal N, Sorek R. 2020. Abortive infection: bacterial suicide as an antiviral immune strategy. *Annu. Rev. Virol.* 7:371–84
20. Akira S, Uematsu S, Takeuchi O. 2006. Pathogen recognition and innate immunity. *Cell* 124:783–801
21. Flajnik MF, Kasahara M. 2010. Origin and evolution of the adaptive immune system: genetic events and selective pressures. *Nat. Rev. Genet.* 11:47–59
22. Hampton HG, Watson BNJ, Fineran PC. 2020. The arms race between bacteria and their phage foes. *Nature* 577:327–36
23. LeRoux M, Laub MT. 2022. Toxin-antitoxin systems as phage defense elements. *Annu. Rev. Microbiol.* 76:21–43
24. Niewoehner O, Garcia-Doval C, Rostol JT, Berk C, Schwede F, et al. 2017. Type III CRISPR-Cas systems produce cyclic oligoadenylate second messengers. *Nature* 548:543–48
25. Kazlauskienė M, Kostiuk G, Venclovas C, Tamulaitis G, Siksnys V. 2017. A cyclic oligonucleotide signaling pathway in type III CRISPR-Cas systems. *Science* 357:605–9
26. Tal N, Morehouse BR, Millman A, Stokar-Avihail A, Avraham C, et al. 2021. Cyclic CMP and cyclic UMP mediate bacterial immunity against phages. *Cell* 184:5728–39
27. Ofir G, Herbst E, Baroz M, Cohen D, Millman A, et al. 2021. Antiviral activity of bacterial TIR domains via immune signalling molecules. *Nature* 600:116–20
28. Leavitt A, Yirmiya E, Amitai G, Lu A, Garb J, et al. 2022. Viruses inhibit TIR gcADPR signalling to overcome bacterial defence. *Nature* 611:326–31

29. Kristiansen H, Gad HH, Eskildsen-Larsen S, Despres P, Hartmann R. 2011. The oligoadenylate synthetase family: an ancient protein family with multiple antiviral activities. *J. Interferon Cytokine Res.* 31:41–47
30. Stetson DB, Medzhitov R. 2006. Recognition of cytosolic DNA activates an IRF3-dependent innate immune response. *Immunity* 24:93–103
31. Eaglesham JB, Kranzusch PJ. 2020. Conserved strategies for pathogen evasion of cGAS-STING immunity. *Curr. Opin. Immunol.* 66:27–34
32. Vashi N, Bakhom SF. 2021. The evolution of STING signaling and its involvement in cancer. *Trends Biochem. Sci.* 46:446–60
33. Hutinet G, Lee YJ, de Crecy-Lagard V, Weigle PR. 2021. Hypermodified DNA in viruses of *E. coli* and *Salmonella*. *EcoSal Plus* 9:eESP00282019
34. Banh DV, Roberts CG, Amador AM, Brady SF, Marraffini LA. 2023. Bacterial cGAS senses a viral RNA to initiate immunity. *bioRxiv* 2023.03.07.531596. <https://doi.org/10.1101/2023.03.07.531596>
35. Huiting E, Cao X, Ren J, Athukoralage JS, Luo Z, et al. 2023. Bacteriophages inhibit and evade cGAS-like immune function in bacteria. *Cell* 186:864–76
36. Gao LA, Wilkinson ME, Strecker J, Makarova KS, Macrae RK, et al. 2022. Prokaryotic innate immunity through pattern recognition of conserved viral proteins. *Science* 377:eabm4096
37. Zhang T, Tamman H, Coppieters T, Wallant K, Kurata T, LeRoux M, et al. 2022. Direct activation of a bacterial innate immune system by a viral capsid protein. *Nature* 612:132–40
38. Duncan-Lowey B, Kranzusch PJ. 2022. CBASS phage defense and evolution of antiviral nucleotide signaling. *Curr. Opin. Immunol.* 74:156–63
39. Kranzusch PJ. 2019. cGAS and CD-NTase enzymes: structure, mechanism, and evolution. *Curr. Opin. Struct. Biol.* 59:178–87
40. Kranzusch PJ, Lee AS, Berger JM, Doudna JA. 2013. Structure of human cGAS reveals a conserved family of second-messenger enzymes in innate immunity. *Cell Rep.* 3:1362–68
41. Civril F, Deimling T, de Oliveira Mann CC, Ablasser A, Moldt M, et al. 2013. Structural mechanism of cytosolic DNA sensing by cGAS. *Nature* 498:332–37
42. Zhang X, Wu J, Du F, Xu H, Sun L, et al. 2014. The cytosolic DNA sensor cGAS forms an oligomeric complex with DNA and undergoes switch-like conformational changes in the activation loop. *Cell Rep.* 6:421–30
43. Li X, Shu C, Yi G, Chaton CT, Shelton CL, et al. 2013. Cyclic GMP-AMP synthase is activated by double-stranded DNA-induced oligomerization. *Immunity* 39:1019–31
44. Zhou W, Whiteley AT, de Oliveira Mann CC, Morehouse BR, Nowak RP, et al. 2018. Structure of the human cGAS-DNA complex reveals enhanced control of immune surveillance. *Cell* 174:300–11
45. Kranzusch PJ, Lee ASY, Wilson SC, Solovych MS, Vance RE, et al. 2014. Structure-guided reprogramming of human cGAS dinucleotide linkage specificity. *Cell* 158:1011–21
46. Zhu D, Wang L, Shang G, Liu X, Zhu J, et al. 2014. Structural biochemistry of a *Vibrio cholerae* dinucleotide cyclase reveals cyclase activity regulation by folates. *Mol. Cell* 55:931–37
47. Andreeva L, Hiller B, Kostrewa D, Lassig C, de Oliveira Mann CC, et al. 2017. cGAS senses long and HMGB/TFAM-bound U-turn DNA by forming protein-DNA ladders. *Nature* 549:394–98
48. Du M, Chen ZJ. 2018. DNA-induced liquid phase condensation of cGAS activates innate immune signaling. *Science* 361:704–9
49. Xie W, Lama L, Adura C, Tomita D, Glickman JF, et al. 2019. Human cGAS catalytic domain has an additional DNA-binding interface that enhances enzymatic activity and liquid-phase condensation. *PNAS* 116:11946–55
50. Zhou W, Mohr L, Maciejowski J, Kranzusch PJ. 2021. cGAS phase separation inhibits TREX1-mediated DNA degradation and enhances cytosolic DNA sensing. *Mol. Cell* 81:739–55
51. Duncan-Lowey B, McNamara-Bordewick NK, Tal N, Sorek R, Kranzusch PJ. 2021. Effector-mediated membrane disruption controls cell death in CBASS antiphage defense. *Mol. Cell* 81:5039–51
52. Burroughs AM, Zhang D, Schaffer DE, Iyer LM, Aravind L. 2015. Comparative genomic analyses reveal a vast, novel network of nucleotide-centric systems in biological conflicts, immunity and signaling. *Nucleic Acids Res.* 43:10633–54

53. Michalski S, de Oliveira Mann CC, Stafford CA, Witte G, Bartho J, et al. 2020. Structural basis for sequestration and autoinhibition of cGAS by chromatin. *Nature* 587:678–82
54. Boyer JA, Spangler CJ, Strauss JD, Cesmat AP, Liu P, et al. 2020. Structural basis of nucleosome-dependent cGAS inhibition. *Science* 370:450–54
55. Cao D, Han X, Fan X, Xu RM, Zhang X. 2020. Structural basis for nucleosome-mediated inhibition of cGAS activity. *Cell Res.* 30:1088–97
56. Kujirai T, Zierhut C, Takizawa Y, Kim R, Negishi L, et al. 2020. Structural basis for the inhibition of cGAS by nucleosomes. *Science* 370:455–58
57. Pathare GR, Decout A, Gluck S, Cavadini S, Makasheva K, et al. 2020. Structural mechanism of cGAS inhibition by the nucleosome. *Nature* 587:668–72
58. Zhao B, Xu P, Rowlett CM, Jing T, Shinde O, et al. 2020. The molecular basis of tight nuclear tethering and inactivation of cGAS. *Nature* 587:673–77
59. Guey B, Wischniewski M, Decout A, Makasheva K, Kaynak M, et al. 2020. BAF restricts cGAS on nuclear DNA to prevent innate immune activation. *Science* 369:823–28
60. Gao D, Li T, Li XD, Chen X, Li QZ, et al. 2015. Activation of cyclic GMP-AMP synthase by self-DNA causes autoimmune diseases. *PNAS* 112:E5699–705
61. Gray EE, Treuting PM, Woodward JJ, Stetson DB. 2015. Cutting edge: cGAS is required for lethal autoimmune disease in the Trex1-deficient mouse model of Aicardi-Goutieres syndrome. *J. Immunol.* 195:1939–43
62. Seo GJ, Kim C, Shin WJ, Sklan EH, Eoh H, Jung JU. 2018. TRIM56-mediated monoubiquitination of cGAS for cytosolic DNA sensing. *Nat. Commun.* 9:613
63. Xia P, Ye B, Wang S, Zhu X, Du Y, et al. 2016. Glutamylation of the DNA sensor cGAS regulates its binding and synthase activity in antiviral immunity. *Nat. Immunol.* 17:369–78
64. Dai J, Huang YJ, He X, Zhao M, Wang X, et al. 2019. Acetylation blocks cGAS activity and inhibits self-DNA-induced autoimmunity. *Cell* 176:1447–60
65. Hu MM, Yang Q, Xie XQ, Liao CY, Lin H, et al. 2016. Sumoylation promotes the stability of the DNA sensor cGAS and the adaptor STING to regulate the kinetics of response to DNA virus. *Immunity* 45:555–69
66. Li T, Huang T, Du M, Chen X, Du F, et al. 2021. Phosphorylation and chromatin tethering prevent cGAS activation during mitosis. *Science* 371:eabc5386
67. Xu G, Liu C, Zhou S, Li Q, Feng Y, et al. 2021. Viral tegument proteins restrict cGAS-DNA phase separation to mediate immune evasion. *Mol. Cell* 81:2823–37
68. Bhowmik D, Du M, Tian Y, Ma S, Wu J, et al. 2021. Cooperative DNA binding mediated by KicGAS/ORF52 oligomerization allows inhibition of DNA-induced phase separation and activation of cGAS. *Nucleic Acids Res.* 49:9389–403
69. Meng F, Yu Z, Zhang D, Chen S, Guan H, et al. 2021. Induced phase separation of mutant NF2 imprisons the cGAS-STING machinery to abrogate antitumor immunity. *Mol. Cell* 81:4147–64
70. Lian H, Wei J, Zang R, Ye W, Yang Q, et al. 2018. ZCCHC3 is a co-sensor of cGAS for dsDNA recognition in innate immune response. *Nat. Commun.* 9:3349
71. Liao CY, Lei CQ, Shu HB. 2021. PCBP1 modulates the innate immune response by facilitating the binding of cGAS to DNA. *Cell Mol. Immunol.* 18:2334–43
72. Tao X, Song J, Song Y, Zhang Y, Yang J, et al. 2022. Ku proteins promote DNA binding and condensation of cyclic GMP-AMP synthase. *Cell Rep.* 40:111310
73. Peters NE, Ferguson BJ, Mazzon M, Fahy AS, Krysztofinska E, et al. 2013. A mechanism for the inhibition of DNA-PK-mediated DNA sensing by a virus. *PLOS Pathog.* 9:e1003649
74. Scutts SR, Ember SW, Ren H, Ye C, Lovejoy CA, et al. 2018. DNA-PK is targeted by multiple vaccinia virus proteins to inhibit DNA sensing. *Cell Rep.* 25:1953–65
75. Rivera-Calzada A, Arribas-Bosacoma R, Ruiz-Ramos A, Escudero-Bravo P, Boskovic J, et al. 2022. Structural basis for the inactivation of cytosolic DNA sensing by the vaccinia virus. *Nat. Commun.* 13:7062
76. Ledvina HE, Ye Q, Gu Y, Sullivan AE, Quan Y, et al. 2023. An E1–E2 fusion protein primes antiviral immune signalling in bacteria. *Nature* 616:319–25

77. Yan Y, Huang F, Yu B, Cheng R, Wu H, et al. 2022. Ubiquitin-like cGAS chain formation by a super enzyme activates anti-phage response. *bioRxiv* 2022.05.25.493364. <https://doi.org/10.1101/2022.05.25.493364>
78. Jenson JM, Li T, Du F, Ea CK, Chen ZJ. 2023. Ubiquitin-like conjugation by bacterial cGAS enhances anti-phage defence. *Nature* 616:326–31
79. Blankenchip CL, Nguyen JV, Lau RK, Ye Q, Gu Y, Corbett KD. 2022. Control of bacterial immune signaling by a WYL domain transcription factor. *Nucleic Acids Res.* 50:5239–50
80. Govande AA, Duncan-Lowey B, Eaglesham JB, Whiteley AT, Kranzusch PJ. 2021. Molecular basis of CD-Ntase nucleotide selection in CBASS anti-phage defense. *Cell Rep.* 35:109206
81. Ko TP, Wang YC, Tsai CL, Yang CS, Hou MH, Chen Y. 2021. Crystal structure and functional implication of a bacterial cyclic AMP-AMP-GMP synthetase. *Nucleic Acids Res.* 49:4725–37
82. Ablasser A, Schmid-Burgk JL, Hemmerling I, Horvath GL, Schmidt T, et al. 2013. Cell intrinsic immunity spreads to bystander cells via the intercellular transfer of cGAMP. *Nature* 503:530–34
83. Luteijn RD, Zaver SA, Gowen BG, Wyman SK, Garelis NE, et al. 2019. SLC19A1 transports immunoreactive cyclic dinucleotides. *Nature* 573:434–38
84. Ritchie C, Cordova AF, Hess GT, Bassik MC, Li L. 2019. SLC19A1 is an importer of the immunotransmitter cGAMP. *Mol. Cell* 75:372–81
85. Lahey LJ, Mardjuki RE, Wen X, Hess GT, Ritchie C, et al. 2020. LRRC8A:C/E heteromeric channels are ubiquitous transporters of cGAMP. *Mol. Cell* 80:578–91
86. Gao P, Asciano M, Zillinger T, Wang W, Dai P, et al. 2013. Structure-function analysis of STING activation by c[G(2',5')pA(3',5')p] and targeting by antiviral DMXAA. *Cell* 154:748–62
87. Kranzusch PJ, Wilson SC, Lee AS, Berger JM, Doudna JA, Vance RE. 2015. Ancient origin of cGAS-STING reveals mechanism of universal 2',3' cGAMP signaling. *Mol. Cell* 59:891–903
88. Davies BW, Bogard RW, Young TS, Mekalanos JJ. 2012. Coordinated regulation of accessory genetic elements produces cyclic di-nucleotides for *V. cholerae* virulence. *Cell* 149:358–70
89. Lau RK, Ye Q, Birkholz EA, Berg KR, Patel L, et al. 2020. Structure and mechanism of a cyclic trinucleotide-activated bacterial endonuclease mediating bacteriophage immunity. *Mol. Cell* 77:723–33
90. Rouillon C, Schneberger N, Chi H, Blumenstock K, Da Vela S, et al. 2022. Antiviral signaling by a cyclic nucleotide activated CRISPR protease. *Nature* 614(7946):168–74
91. McFarland AP, Luo S, Ahmed-Qadri F, Zuck M, Thayer EF, et al. 2017. Sensing of bacterial cyclic dinucleotides by the oxidoreductase RECON promotes NF- κ B activation and shapes a proinflammatory antibacterial state. *Immunity* 46:433–45
92. Fatma S, Chakravarti A, Zeng X, Huang RH. 2021. Molecular mechanisms of the CdnG-Cap5 antiphage defense system employing 3',2'-cGAMP as the second messenger. *Nat. Commun.* 12:6381
93. Maltback JH, Cambier S, Snyder JM, Stetson DB. 2022. ABCC1 transporter exports the immunostimulatory cyclic dinucleotide cGAMP. *Immunity* 55:1799–812
94. Marcus A, Mao AJ, Lensink-Vasan M, Wang L, Vance RE, Raulet DH. 2018. Tumor-derived cGAMP triggers a STING-mediated interferon response in non-tumor cells to activate the NK cell response. *Immunity* 49:754–63
95. Carozza JA, Cordova AF, Brown JA, AlSaif Y, Bohnert V, et al. 2022. ENPP1's regulation of extracellular cGAMP is a ubiquitous mechanism of attenuating STING signaling. *PNAS* 119:e2119189119
96. Li L, Yin Q, Kuss P, Maliga Z, Millan JL, et al. 2014. Hydrolysis of 2'3'-cGAMP by ENPP1 and design of nonhydrolyzable analogs. *Nat. Chem. Biol.* 10:1043–48
97. Bridgeman A, Maelfait J, Davenne T, Partridge T, Peng Y, et al. 2015. Viruses transfer the antiviral second messenger cGAMP between cells. *Science* 349:1228–32
98. Gentili M, Kowal J, Tkach M, Satoh T, Lahaye X, et al. 2015. Transmission of innate immune signaling by packaging of cGAMP in viral particles. *Science* 349:1232–36
99. Segrist E, Dittmar M, Gold B, Cherry S. 2021. Orally acquired cyclic dinucleotides drive dSTING-dependent antiviral immunity in enterocytes. *Cell Rep.* 37:110150
100. Eaglesham JB, Pan Y, Kupper TS, Kranzusch PJ. 2019. Viral and metazoan poxins are cGAMP-specific nucleases that restrict cGAS-STING signalling. *Nature* 566:259–63
101. Hobbs SJ, Wein T, Lu A, Morehouse BR, Schnabel J, et al. 2022. Phage anti-CBASS and anti-Pycsar nucleases subvert bacterial immunity. *Nature* 605:522–26

102. Maluquer de Motes C. 2021. Poxvirus cGAMP nucleases: clues and mysteries from a stolen gene. *PLOS Pathog.* 17:e1009372
103. Eaglesham JB, McCarty KL, Kranzusch PJ. 2020. Structures of diverse poxins cGAMP nucleases reveal a widespread role for cGAS-STING evasion in host-pathogen conflict. *eLife* 9:e59753
104. Hernaez B, Alonso G, Georgana I, El-Jesr M, Martin R, et al. 2020. Viral cGAMP nuclease reveals the essential role of DNA sensing in protection against acute lethal virus infection. *Sci. Adv.* 6(38):eabb4565
105. Burdette DL, Monroe KM, Sotelo-Troha K, Iwig JS, Eckert B, et al. 2011. STING is a direct innate immune sensor of cyclic di-GMP. *Nature* 478:515–18
106. Shang G, Zhang C, Chen ZJ, Bai XC, Zhang X. 2019. Cryo-EM structures of STING reveal its mechanism of activation by cyclic GMP-AMP. *Nature* 567:389–93
107. de Oliveira Mann CC, Orzalli MH, King DS, Kagan JC, Lee ASY, Kranzusch PJ. 2019. Modular architecture of the STING C-terminal tail allows interferon and NF- κ B signaling adaptation. *Cell Rep.* 27:1165–75
108. Wu X, Wu FH, Wang X, Wang L, Siedow JN, et al. 2014. Molecular evolutionary and structural analysis of the cytosolic DNA sensor cGAS and STING. *Nucleic Acids Res.* 42:8243–57
109. Morehouse BR, Yip MCJ, Keszei AFA, McNamara-Bordewick NK, Shao S, Kranzusch PJ. 2022. Cryo-EM structure of an active bacterial TIR-STING filament complex. *Nature* 608:803–7
110. Hogrel G, Guild A, Graham S, Rickman H, Gruschow S, et al. 2022. Cyclic nucleotide-induced helical structure activates a TIR immune effector. *Nature* 608:808–12
111. Huang YH, Liu XY, Du XX, Jiang ZF, Su XD. 2012. The structural basis for the sensing and binding of cyclic di-GMP by STING. *Nat. Struct. Mol. Biol.* 19:728–30
112. Ouyang S, Song X, Wang Y, Ru H, Shaw N, et al. 2012. Structural analysis of the STING adaptor protein reveals a hydrophobic dimer interface and mode of cyclic di-GMP binding. *Immunity* 36:1073–86
113. Dey B, Dey RJ, Cheung LS, Pokkali S, Guo H, et al. 2015. A bacterial cyclic dinucleotide activates the cytosolic surveillance pathway and mediates innate resistance to tuberculosis. *Nat. Med.* 21:401–6
114. Sixt BS, Bastidas RJ, Finethy R, Baxter RM, Carpenter VK, et al. 2017. The *Chlamydia trachomatis* inclusion membrane protein CpoS counteracts STING-mediated cellular surveillance and suicide programs. *Cell Host Microbe* 21:113–21
115. Woodward JJ, Iavarone AT, Portnoy DA. 2010. c-di-AMP secreted by intracellular *Listeria monocytogenes* activates a host type I interferon response. *Science* 328:1703–5
116. Erttmann SF, Swacha P, Aung KM, Brindelfalk B, Jiang H, et al. 2022. The gut microbiota prime systemic antiviral immunity via the cGAS-STING-IFN-I axis. *Immunity* 55:847–61
117. Lam KC, Araya RE, Huang A, Chen Q, Di Modica M, et al. 2021. Microbiota triggers STING-type I IFN-dependent monocyte reprogramming of the tumor microenvironment. *Cell* 184:5338–56
118. Ko TP, Wang YC, Yang CS, Hou MH, Chen CJ, et al. 2022. Crystal structure and functional implication of bacterial STING. *Nat. Commun.* 13:26
119. Severin GB, Ramliden MS, Hawver LA, Wang K, Pell ME, et al. 2018. Direct activation of a phospholipase by cyclic GMP-AMP in El Tor *Vibrio cholerae*. *PNAS* 115:E6048–55
120. Zhang C, Shang G, Gui X, Zhang X, Bai XC, Chen ZJ. 2019. Structural basis of STING binding with and phosphorylation by TBK1. *Nature* 567:394–98
121. Chu TT, Tu X, Yang K, Wu J, Repa JJ, Yan N. 2021. Tonic prime-boost of STING signalling mediates Niemann-Pick disease type C. *Nature* 596:570–75
122. Liu Y, Xu P, Rivara S, Liu C, Ricci J, et al. 2022. Clathrin-associated AP-1 controls termination of STING signalling. *Nature* 610:761–67
123. Luo WW, Li S, Li C, Lian H, Yang Q, et al. 2016. iRhom2 is essential for innate immunity to DNA viruses by mediating trafficking and stability of the adaptor STING. *Nat. Immunol.* 17:1057–66
124. Zhao B, Du F, Xu P, Shu C, Sankaran B, et al. 2019. A conserved PLPLRT/SD motif of STING mediates the recruitment and activation of TBK1. *Nature* 569:718–22
125. Liu S, Cai X, Wu J, Cong Q, Chen X, et al. 2015. Phosphorylation of innate immune adaptor proteins MAVS, STING, and TRIF induces IRF3 activation. *Science* 347:aaa2630
126. Zhao B, Shu C, Gao X, Sankaran B, Du F, et al. 2016. Structural basis for concerted recruitment and activation of IRF-3 by innate immune adaptor proteins. *PNAS* 113:E3403–12

127. Chen H, Sun H, You F, Sun W, Zhou X, et al. 2011. Activation of STAT6 by STING is critical for antiviral innate immunity. *Cell* 147:436–46
128. McWhirter SM, Barbalat R, Monroe KM, Fontana MF, Hyodo M, et al. 2009. A host type I interferon response is induced by cytosolic sensing of the bacterial second messenger cyclic-di-GMP. *J. Exp. Med.* 206:1899–911
129. Wu J, Yan N. 2022. No longer a one-trick pony: STING signaling activity beyond interferon. *J. Mol. Biol.* 434:167257
130. Goto A, Okado K, Martins N, Cai H, Barbier V, et al. 2018. The kinase IKK β regulates a STING- and NF- κ B-dependent antiviral response pathway in *Drosophila*. *Immunity* 49:225–34
131. Liu Y, Gordesky-Gold B, Leney-Greene M, Weinbren NL, Tudor M, Cherry S. 2018. Inflammation-induced, STING-dependent autophagy restricts Zika virus infection in the *Drosophila* brain. *Cell Host Microbe* 24:57–68
132. Martin M, Hiroyasu A, Guzman RM, Roberts SA, Goodman AG. 2018. Analysis of *Drosophila* STING reveals an evolutionarily conserved antimicrobial function. *Cell Rep.* 23:3537–50
133. Cai H, Holleufer A, Simonsen B, Schneider J, Lemoine A, et al. 2020. 2'3'-cGAMP triggers a STING- and NF- κ B-dependent broad antiviral response in *Drosophila*. *Sci. Signal.* 13(660):eabc4537
134. Gui X, Yang H, Li T, Tan X, Shi P, et al. 2019. Autophagy induction via STING trafficking is a primordial function of the cGAS pathway. *Nature* 567:262–66
135. Margolis SR, Dietzen PA, Hayes BM, Wilson SC, Remick BC, et al. 2021. The cyclic dinucleotide 2'3'-cGAMP induces a broad antibacterial and antiviral response in the sea anemone *Nematostella vectensis*. *PNAS* 118:e2109022118
136. Woznica A, Kumar A, Sturge CR, Xing C, King N, Pfeiffer JK. 2021. STING mediates immune responses in the closest living relatives of animals. *eLife* 10:e70436
137. Hayden MS, Ghosh S. 2011. NF- κ B in immunobiology. *Cell Res.* 21:223–44
138. Zhang T, Ma C, Zhang Z, Zhang H, Hu H. 2021. NF- κ B signaling in inflammation and cancer. *MedComm* 2020 2:618–53
139. Schoggins JW. 2019. Interferon-stimulated genes: What do they all do? *Annu. Rev. Virol.* 6:567–84
140. Bernheim A, Millman A, Ofir G, Meitav G, Avraham C, et al. 2021. Prokaryotic viperins produce diverse antiviral molecules. *Nature* 589:120–24
141. Johnson AG, Wein T, Mayer ML, Duncan-Lowey B, Yirmiya E, et al. 2022. Bacterial gasdermins reveal an ancient mechanism of cell death. *Science* 375:221–25
142. Bernheim A, Sorek R. 2020. The pan-immune system of bacteria: antiviral defence as a community resource. *Nat. Rev. Microbiol.* 18:113–19
143. Gao L, Altae-Tran H, Bohning F, Makarova KS, Segel M, et al. 2020. Diverse enzymatic activities mediate antiviral immunity in prokaryotes. *Science* 369:1077–84
144. Millman A, Melamed S, Leavitt A, Doron S, Bernheim A, et al. 2022. An expanded arsenal of immune systems that protect bacteria from phages. *Cell Host Microbe* 30:1556–69
145. Gruschow S, Adamson CS, White MF. 2021. Specificity and sensitivity of an RNA targeting type III CRISPR complex coupled with a NucC endonuclease effector. *Nucleic Acids Res.* 49:13122–34
146. Mayo-Munoz D, Smith LM, Garcia-Doval C, Malone LM, Harding KR, et al. 2022. Type III CRISPR-Cas provides resistance against nucleus-forming jumbo phages via abortive infection. *Mol. Cell* 82:4471–86
147. Essuman K, Summers DW, Sasaki Y, Mao X, Yim AKY, et al. 2018. TIR domain proteins are an ancient family of NAD⁺-consuming enzymes. *Curr. Biol.* 28:421–30
148. Horsefield S, Burdett H, Zhang X, Manik MK, Shi Y, et al. 2019. NAD⁺ cleavage activity by animal and plant TIR domains in cell death pathways. *Science* 365:793–99
149. Garb J, Lopatina A, Bernheim A, Zaremba M, Siksnys V, et al. 2022. Multiple phage resistance systems inhibit infection via SIR2-dependent NAD⁺ depletion. *Nat. Microbiol.* 7:1849–56
150. Rousset F, Yirmiya E, Neshet S, Brandis A, Mehlman T, et al. 2023. A conserved family of immune effectors cleaves cellular ATP upon viral infection. *bioRxiv* 2023.01.24.525353. <https://doi.org/10.1101/2023.01.24.525353>
151. Hee CS, Habazettl J, Schmutz C, Schirmer T, Jenal U, Grzesiek S. 2020. Intercepting second-messenger signaling by rationally designed peptides sequestering c-di-GMP. *PNAS* 117:17211–20

152. VanderWal AR, Park J-U, Polevoda B, Nicosia JK, Molina Vargas AM, et al. 2023. Csx28 is a membrane pore that enhances CRISPR-Cas13b-dependent antiphage defense. *Science* 380(6643):410–15
153. Doron S, Melamed S, Ofir G, Leavitt A, Lopatina A, et al. 2018. Systematic discovery of antiphage defense systems in the microbial pangenome. *Science* 359:eaar4120
154. Makarova KS, Wolf YI, Snir S, Koonin EV. 2011. Defense islands in bacterial and archaeal genomes and prediction of novel defense systems. *J. Bacteriol.* 193:6039–56
155. Johnson MC, Laderman E, Huiting E, Zhang C, Davidson A, Bondy-Denomy J. 2022. Core defense hotspots within *Pseudomonas aeruginosa* are a consistent and rich source of anti-phage defense systems. bioRxiv 2022.11.11.516204. <https://doi.org/10.1101/2022.11.11.516204>
156. Koonin EV, Makarova KS, Wolf YI. 2017. Evolutionary genomics of defense systems in archaea and bacteria. *Annu. Rev. Microbiol.* 71:233–61
157. Martin WF, Garg S, Zimorski V. 2015. Endosymbiotic theories for eukaryote origin. *Philos. Trans. R. Soc. B* 370:20140330
158. Wein T, Sorek R. 2022. Bacterial origins of human cell-autonomous innate immune mechanisms. *Nat. Rev. Immunol.* 22:629–38
159. Cury J, Mordret E, Trejo VH, Tesson F, Ofir G, et al. 2022. Conservation of antiviral systems across domains of life reveals novel immune mechanisms in humans. bioRxiv 2022.12.12.520048. <https://doi.org/10.1101/2022.12.12.520048>
160. Chou S, Daugherty MD, Peterson SB, Biboy J, Yang Y, et al. 2015. Transferred interbacterial antagonism genes augment eukaryotic innate immune function. *Nature* 518:98–101
161. Dunning Hotopp JC. 2011. Horizontal gene transfer between bacteria and animals. *Trends Genet.* 27:157–63
162. de Oliveira Mann CC, Kiefersauer R, Witte G, Hopfner KP. 2016. Structural and biochemical characterization of the cell fate determining nucleotidyltransferase fold protein MAB21L1. *Sci. Rep.* 6:27498
163. Horn D, Prescott T, Houge G, Braekke K, Rosendahl K, et al. 2015. A Novel Oculo-Skeletal syndrome with intellectual disability caused by a particular *MAB21L2* mutation. *Eur. J. Med. Genet.* 58:387–91
164. Hovanessian AG, Brown RE, Kerr IM. 1977. Synthesis of low molecular weight inhibitor of protein synthesis with enzyme from interferon-treated cells. *Nature* 268:537–40
165. Kerr IM, Brown RE. 1978. pppA2'p5'A2'p5'A: an inhibitor of protein synthesis synthesized with an enzyme fraction from interferon-treated cells. *PNAS* 75:256–60